



ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your *Hashimoto's*, Ultimate Guide

A Complete Functional Medicine Guide to Calming the Autoimmune Attack, Addressing the Root Triggers, and Reclaiming Your Thyroid Health

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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The Medication Is Not Enough

There is a conversation that happens in functional medicine practices every week, repeated in thousands of variations but always carrying the same essential frustration. A patient sits down and explains that they have been on thyroid medication for two, five, sometimes ten years. Their TSH is controlled. Their endocrinologist is satisfied. But they are exhausted in a way that sleep does not touch. They cannot lose weight despite genuine effort. Their hair continues to thin. Their brain fog is worse than ever. Their mood is unpredictable. They feel nothing like themselves.

The medication is doing exactly what it was prescribed to do. It is replacing the thyroid hormone that the gland is no longer producing adequately. What it is not doing, and was never designed to do, is stop the immune system from attacking the thyroid. In Hashimoto's thyroiditis, the root problem is the immune system, not the thyroid. Treating the thyroid in isolation while leaving the immune attack unaddressed addresses the hormone deficiency but not the process driving it.

Hashimoto's thyroiditis is the most common autoimmune disease in the United States and the primary cause of hypothyroidism in developed countries.¹ Despite its prevalence, the majority of patients who carry the diagnosis have never been told what drives the autoimmune attack, what maintains it, or what specifically can be done to reduce it. They have been given a medication and told their labs are managed. The progressive destruction of their thyroid tissue continues.

This guide takes a different position. The autoimmune attack in Hashimoto's is not random and it is not irreversible in all patients. It has specific, identifiable, and addressable triggers. Gut permeability allows antigen traffic that activates and sustains the immune response. Gluten cross-reactivity produces antibodies that attack both intestinal tissue and thyroid tissue through molecular mimicry. A common foodborne pathogen called *Yersinia enterocolitica* produces an outer membrane protein associated with cross-reactive immunity against the TSH receptor. *Helicobacter pylori* gastric colonization drives both molecular mimicry and secondary gut permeability that amplifies the autoimmune signal. Gut dysbiosis creates an LPS-driven inflammatory environment that sustains Th17 polarization. Lyme disease drives the same immune polarization that characterizes Hashimoto's lymphocytic thyroid infiltration. Nutrient deficiencies impair the enzyme systems that govern thyroid hormone production and conversion.

Every one of these drivers is testable. Every one is addressable. In my clinical experience, the patients who recover their energy and cognitive function are the ones whose care addresses these upstream factors rather than only the downstream hormone deficiency.

One correction to make before you read another page, because it governs everything that follows.

You have almost certainly been told that your antibodies matter and that lowering them is the goal. Antibody titers are the most-measured, most-discussed, and least outcome-validated number in this condition. Several interventions genuinely lower them. What nobody has demonstrated is that lowering them makes

you feel better or changes where your disease goes. They are still worth watching. They are just not a scoreboard, and you should be suspicious of anyone, including me, whose success metric is a falling TPO number. Throughout this book I use antibodies to establish that the process is autoimmune and to monitor it. I do not use them as the outcome, and neither should you. The outcomes worth tracking are energy, cognition, sleep, hair, cold tolerance, and how you feel by mid-afternoon.

What you are buying here, stated plainly. The mechanisms in this book are drawn from immunology and from association studies, and they are largely sound as mechanisms. What is thin, almost everywhere in this field, is outcome evidence: trials showing that acting on a given driver changes how a Hashimoto's patient feels or where the disease goes. Selenium and vitamin D have antibody data of varying quality and essentially no quality-of-life data. The infectious triggers have association data and, in the case of H. pylori, eradication data. The rest is reasoned clinical practice held to an endpoint. That is what happens when nobody can patent the intervention: the trials never get funded. So read this as a framework for investigation rather than as proven protocol, put your money in the order the evidence sits, and hold everything you start to a defined endpoint.

This guide walks you through eight keys. Read it as a clinical framework, not as a supplement shopping list. The goal is understanding, and understanding is what makes every subsequent clinical decision more effective.

What Hashimoto's Actually Is

The distinction between hypothyroidism and Hashimoto's thyroiditis matters clinically, because it separates a treatment plan focused only on hormone replacement from one that also addresses the immune process driving the destruction. Understanding what Hashimoto's actually is, at the mechanistic level, is the foundation on which everything else in this guide rests.

Hypothyroidism vs. Hashimoto's

Hypothyroidism is a state: insufficient thyroid hormone production. Hashimoto's thyroiditis is a cause: an autoimmune disease in which the immune system generates antibodies against thyroid tissue, progressively destroying the gland's functional capacity. Most cases of hypothyroidism in the United States are caused by Hashimoto's, but the reverse is not true. Not all Hashimoto's patients are hypothyroid, at least not at diagnosis. Early in the disease process, the inflammatory destruction can actually cause transient hyperthyroid episodes as stored thyroid hormone leaks from damaged tissue into circulation. Over time, as destruction progresses, the gland loses functional mass and hypothyroidism becomes established.

This distinction matters because it determines what you investigate. For a physician focused on hypothyroidism, a controlled TSH is the endpoint. For a clinician addressing Hashimoto's, the controlled TSH is where the work starts rather than where it stops, because hormone replacement addresses the hormone deficiency and does nothing to the immune process, the gut, the nutrient status, or the load. That is the reason people can be perfectly medicated and still feel unwell, and it is why this book exists. Note what I am not saying: I am not saying a persistently high antibody number means treatment has failed. Antibodies establish that the process is autoimmune and are worth monitoring; they have not been shown to track how you feel or where you are heading.

The Autoimmune Mechanism

The immune system maintains a critical distinction between self and non-self. Tolerance to self-antigens is established during immune development through a process of eliminating or suppressing immune cells that react to the body's own proteins. When this tolerance breaks down, the immune system generates antibodies and cytotoxic cells that attack the body's own tissues. This is the definition of autoimmunity.

In Hashimoto's, the primary antibody targets are thyroid peroxidase (TPO), the enzyme essential for thyroid hormone synthesis, and thyroglobulin, the protein matrix in which thyroid hormones are stored. When the immune system produces TPO antibodies (TPOAb) and thyroglobulin antibodies (TgAb), it recruits lymphocytes into the thyroid gland in a process called lymphocytic infiltration. These lymphocytes release cytokines that drive chronic inflammation within the gland. Over years, the inflammatory destruction reduces the number of functioning follicular cells, and thyroid hormone output declines.

This process can be active for years or decades before TSH rises above the conventional laboratory cutoff for hypothyroidism. Patients with significant antibody elevations and progressive thyroid destruction may have entirely normal TSH results during this period. This is why antibody testing is not optional in a thyroid workup and why a normal TSH does not rule out active Hashimoto's disease.

The Thyroid-Immune Bidirectional Loop

One of the less-discussed aspects of Hashimoto's is that the destruction of thyroid tissue itself perpetuates the autoimmune response. As thyroid cells are damaged and lysed by lymphocytic attack, their contents, including TPO and thyroglobulin, are released into circulation. This ongoing antigen exposure continuously stimulates the immune system to produce more antibodies. The process is self-reinforcing. This is the mechanism that explains why, without addressing the upstream triggers that initiated the immune response, the autoimmune attack tends to progress over time rather than spontaneously resolving.

Addressing Hashimoto's clinically means two things simultaneously: reducing the upstream drivers that are activating and sustaining the immune response, and supporting the nutritional and metabolic conditions the immune system depends on. Neither alone is sufficient. What that combination is aimed at is how you function, not a titer. Judge it there.

The Complete Testing Framework

The standard thyroid test, a TSH drawn at an annual physical, is the beginning of a workup for Hashimoto's, not the end. Understanding what each marker reveals, and which markers are routinely omitted that should not be, is essential for any patient who wants to understand the full picture of their autoimmune thyroid status.

The Core Thyroid Panel

TSH, or thyroid-stimulating hormone, is produced by the pituitary gland and reflects the brain's assessment of circulating thyroid hormone levels. It is a valuable screening marker, but its limitations are significant in the context of Hashimoto's. TSH can remain in the conventional normal range (0.4 to 4.5 mIU/L) for years while active autoimmune destruction is progressing. The functional target I work toward, 1.0 to 2.5 mIU/L, is considerably tighter and more clinically meaningful for patients with active Hashimoto's. It is the same target printed in the lab interpretation guide in this kit, and it is a clinical target used to guide investigation rather than a validated diagnostic threshold.

Free T4 is the primary hormone produced by the thyroid gland. It is inactive until converted to the active T3 form. Free T3 is the active thyroid hormone that binds to receptors on virtually every cell in the body. Low Free T3, even with normal TSH and normal Free T4, produces the full clinical picture of hypothyroidism at the cellular level and is among the most common explanations for persistent symptoms in medicated Hashimoto's patients. The conversion of T4 to T3 depends on selenium, zinc, adequate liver function, and low inflammatory burden, all of which are commonly compromised in Hashimoto's.

Reverse T3 is an inactive metabolite of T4 that competes with active T3 for receptor binding. When the body is under chronic inflammatory stress, as it is in active Hashimoto's, T4 is preferentially converted to Reverse T3 rather than active T3. Elevated Reverse T3 produces cellular hypothyroidism even when circulating hormone levels appear adequate. The Free T3 to Reverse T3 ratio is one of the most useful single numbers in a Hashimoto's workup.

TPO antibodies and thyroglobulin antibodies are the markers that establish the process is autoimmune rather than something else. That is diagnostically useful, and positive antibodies alongside a normal TSH also predict a higher likelihood of developing hypothyroidism later, which justifies monitoring. What they are not is a treatment target. Several interventions lower them, and none has been shown to make anyone feel better by doing so, so I do not chase them toward undetectable and I would treat any protocol that judges itself on your falling TPO number with suspicion: it has chosen the outcome that is easier to move, not the one you came for. The lab guide in this kit takes this apart in full.

The Extended Hashimoto's Workup

Beyond the core thyroid panel, a complete Hashimoto's workup at Root Health includes several additional assessments that address the upstream drivers of the autoimmune process.

Test	What It Reveals	Clinical Significance
Vibrant Hormone Zoomer	Full thyroid panel (TSH, Free T3, Free T4, Reverse T3, TPOAb, TgAb), adrenal cortisol rhythm, comprehensive sex hormones	Single draw captures thyroid conversion, autoimmune activity, and adrenal status.
Vibrant Wheat Zoomer	Tight junction antibodies (occludin/zonulin), gluten and wheat peptide reactivity across 26 antigens	Suggestive rather than diagnostic for gut permeability and gluten reactivity. Does not replace celiac serology.
LabCorp H. pylori Stool Antigen or Urea Breath Test	Active H. pylori gastric colonization. Stool antigen or urea breath test are preferred over serology for detection of active infection.	Inexpensive, high-yield first-line screen. Eradication studies show measurable TPO antibody reduction. Order before the expensive panels.
LabCorp Yersinia Serology (IgG/IgA/IgM)	Antibodies to Yersinia enterocolitica, a foodborne pathogen associated with thyroid autoimmunity in cross-sectional research.	Consider in non-responders after initial gut and dietary intervention. Not on any standard thyroid or tick-borne panel. Order through the office.
Vibrant Tickborne 2.0 PCR	Borrelia burgdorferi (41 antigens), Babesia, Bartonella, Ehrlichia, Anaplasma	For patients in tick-endemic areas or with non-resolving antibodies after gut and dietary intervention. The most expensive panel on this list, which is why it comes last.
LabCorp EBV Panel (VCA IgG/IgM, EA IgG, EBNA IgG)	Distinguishes past resolved EBV infection, active reactivation (EA positive + VCA IgG positive + EBNA positive), and recent primary infection (VCA IgM positive)	EBV is a plausible driver rather than a certainty, and there is no validated antiviral protocol to act on a positive with. Add when fatigue, lymph node history, or immune dysregulation are prominent.
Vibrant Gut Zoomer Complete	Microbiome composition, pathogen detection, permeability markers, secretory IgA, short-chain fatty acids, Blastocystis hominis	Essential when gut symptoms are present, when antibodies are not responding, or to characterize dysbiosis before antimicrobial intervention.
Vibrant Micronutrient Panel	Selenium, zinc, iodine status, B vitamins, vitamin D, magnesium, iron	Nutrient deficiencies impair T4-to-T3 conversion. Ferritin, B12, folate, zinc, and vitamin D can also be run individually for a fraction of the cost, and often should be.

The order of investigation should be guided by the clinical picture. For all confirmed Hashimoto's patients, the Hormone Zoomer and H. pylori stool antigen are the standard first-line tests, providing the highest yield at the lowest cost. Yersinia serology is added for patients who are not responding to initial gut and dietary

intervention. Wheat Zoomer is added when gluten reactivity is suspected. Tickborne 2.0 PCR is added for patients with tick exposure history or antibodies not responding after gut and infectious workup.

The Gut–Thyroid Connection

The relationship between gut health and thyroid autoimmunity is mechanistic and specific, and it is addressable. Understanding the gut-thyroid pathways in Hashimoto's changes the entire treatment framework from one focused only on the thyroid to one that treats the upstream drivers where the autoimmune process actually begins.

Pathway One: Intestinal Permeability

The gut barrier, when functioning correctly, allows properly digested nutrients to enter circulation while blocking undigested proteins, bacterial endotoxins, and pathogenic organisms. When tight junction proteins are disrupted, this barrier becomes permeable, and substances that should remain in the gut lumen gain access to systemic circulation. The immune system encounters these foreign antigens and mounts a response.

In Hashimoto's, the specific consequence of this increased antigen exposure is activation of the autoimmune cascade through a process called bystander activation, in which immune cells activated against gut-derived antigens generate cross-reactive responses against thyroid tissue.² The persistent antigen traffic through a leaky gut maintains a chronic state of immune activation that keeps thyroid antibody production elevated even when other triggers are addressed.

This is the reasoning behind adding gut barrier repair when dietary change alone has not helped. One caution on how far to push that reasoning: a 2025 study found intestinal permeability to be a modifiable contributor in Hashimoto's with causality unresolved.² “Modifiable contributor with causality unresolved” is a fair reason to work on the gut. It is not a promise that fixing the gut resolves the disease, and read any permeability panel as a pattern rather than as proof from a single marker.

Pathway Two: Gluten and Molecular Mimicry

Gliadin, the immunogenic fraction of gluten, is proposed to share structural epitopes with thyroid tissue, so that in patients carrying HLA-DQ2 or HLA-DQ8 the immune response to gliadin generates antibodies that cross-react with thyroid antigens. That is molecular mimicry, it is a coherent immunological hypothesis, and the effect of gluten-free diets in non-celiac Hashimoto's has been examined in a systematic review and meta-analysis.³

Here is where I part company with most thyroid content, though I am not dismissing gluten. The standard advice is: you have Hashimoto's, go gluten free. I think that advice has the order of operations backwards, and the diet guide in this kit lays out why in full, including a head-to-head trial and a Mendelian randomization analysis. The short version is that in celiac disease the pathogenic role of gluten is firmly

established, while in Hashimoto's gluten is better viewed as a **candidate modifier rather than a proven universal trigger**.

So here is the sequence I actually use.

Get tested for celiac disease first, while you are still eating gluten. Hashimoto's raises your celiac risk measurably. Celiac is a real diagnosis with real testing and a treatment that genuinely works, and that testing depends on you currently eating gluten. Go gluten free first and you lose the ability to get a clean answer without a deliberate gluten challenge later. This is the single most valuable action in the whole dietary section, and skipping it is the most common expensive mistake I see.

If celiac testing is positive, gluten free is the treatment. Not a wellness experiment. A treatment, and strict is the right word there, because cross-contamination from shared surfaces, wheat starch in processed foods, and soy sauce all count.

If celiac testing is negative and you still want to try it, that is reasonable. Run it as a time-limited experiment with a defined endpoint, eight to twelve weeks, judged on how you feel rather than on a repeat antibody panel.

The Wheat Zoomer panel tests reactivity to 26 gluten and wheat peptides beyond the standard celiac antigens, which can be informative when celiac serology is negative and the question is still open. Read it as suggestive rather than diagnostic. It does not replace celiac serology and it does not establish that gluten is driving your thyroid disease.

Pathway Three: Yersinia Enterocolitica

Yersinia enterocolitica is a gram-negative foodborne bacterium transmitted through contaminated pork, unpasteurized dairy, and undercooked meats. It is not a tick-borne organism, and no standard thyroid panel covers it. Its association with Hashimoto's thyroiditis is one of the more studied infectious connections in thyroid autoimmunity research, with multiple cross-sectional studies documenting significantly elevated *Yersinia* antibodies in Hashimoto's patient populations compared to healthy controls.

The proposed mechanism involves a low molecular weight envelope protein (LP) produced by *Yersinia* that shares structural homology with the human TSH receptor.⁴ The hypothesis is that antibodies generated against this protein during *Yersinia* infection may cross-react with TSH receptor epitopes through molecular mimicry. This mechanism is biologically plausible and supported by the cross-sectional association data. A prospective cohort study (Efthymiou et al., 2011) did not find *Yersinia* seropositivity predictive of de novo thyroid autoimmunity development, which means the causal relationship has not been confirmed and the association may reflect shared genetic susceptibility.⁵ The clinical position I take at Root Health is that *Yersinia* is worth ruling out in patients who are not responding to dietary and gut barrier intervention, because identified *Yersinia* burden is addressable and the test is inexpensive.

Testing is a standalone LabCorp order for *Yersinia enterocolitica* IgG, IgA, and IgM serology. Elevated IgG indicates past or chronic exposure. A raised IgA suggests active mucosal infection, and a raised IgM

indicates recent or acute infection. The infection can be subclinical, producing no gastrointestinal symptoms while maintaining an antigenic burden that keeps thyroid antibodies elevated.

Clinical Note: *Elevated TPO antibodies that do not normalize despite dietary intervention and gut repair are a strong clinical signal that an upstream infectious trigger is maintaining the immune activation. H. pylori and Yersinia are the two most cost-effective investigations at that point. Do not stop investigating at the gut barrier.*

Pathway Four: H. pylori and Gastric Immune Activation

Helicobacter pylori is a gram-negative bacterium that colonizes the gastric mucosa in approximately 44 percent of the global population, making it one of the most prevalent chronic bacterial infections in the world. Its connection to Hashimoto's thyroiditis is supported by multiple cross-sectional studies showing significantly elevated H. pylori seroprevalence in Hashimoto's patients compared to healthy controls, and by eradication studies demonstrating measurable reductions in TPO antibody levels following successful treatment.

The mechanisms are two-pronged. First, the H. pylori CagA protein (cytotoxin-associated gene A) shares structural similarities with thyroid antigens, generating cross-reactive antibodies through molecular mimicry.⁶ Second, H. pylori colonization produces chronic gastric inflammation that progressively impairs stomach acid production. Adequate gastric acid is required for protein denaturation in the stomach. When acid production is insufficient, intact food proteins reach the small intestine without proper denaturation, increasing the antigen load that reaches the gut-associated lymphoid tissue and promoting the permeability-driven immune activation described in Pathway One.

The clinical significance of H. pylori in Hashimoto's management is substantial because eradication data shows actual outcome benefit. Unlike the Yersinia association, which has cross-sectional support but limited prospective outcome data, H. pylori eradication studies have shown direct antibody reduction in Hashimoto's patients following successful treatment. This makes H. pylori one of the highest-yield investigations in Hashimoto's patients who are not responding to dietary and gut barrier intervention. Testing is inexpensive: a LabCorp stool antigen or urea breath test detects active infection with high sensitivity. Serology is less preferred because it cannot distinguish active infection from past exposure.

There is a second reason to care about H. pylori here, and it has nothing to do with autoimmunity. Levothyroxine needs stomach acid to dissolve before it can be absorbed, and H. pylori gastritis lowers acid production. In a New England Journal of Medicine study, patients with H. pylori-related gastritis, atrophic gastritis, or both required 22 to 34 percent more thyroxine than a reference group, and eradicating the organism largely reversed the effect.²³ So in someone whose TSH will not settle despite dose increases, H. pylori may be driving the autoimmune process and blocking the treatment for it at the same time. The lab guide in this kit covers what to do about that.

Pathway Five: Gut Dysbiosis and the LPS-Autoimmunity Loop

Gut dysbiosis, an imbalance in the microbial ecosystem characterized by reduced diversity, depletion of beneficial short-chain fatty acid producers, and relative overgrowth of pro-inflammatory gram-negative species, creates a systemic inflammatory environment that sustains Hashimoto's autoimmunity through a mechanism distinct from gut permeability itself. The two pathways interact, but understanding them separately improves the precision of clinical intervention.

The primary mechanism is LPS endotoxemia. Lipopolysaccharide, the outer membrane component of gram-negative bacteria, ranks among the most potent activators of the innate immune system via Toll-like receptor 4 signaling. In a healthy microbiome with adequate gut barrier function, LPS is contained within the intestinal lumen. When dysbiosis increases the gram-negative bacterial load and gut permeability allows LPS translocation into systemic circulation, TLR-4 activation drives a pro-inflammatory cytokine cascade that sustains the Th17-dominant immune environment characteristic of Hashimoto's.⁷ The autoimmune thyroid attack is, at its base, a Th17-mediated process. Persistent LPS signaling from a dysbiotic gut is a consistent upstream driver of that polarization.

Short-chain fatty acids, particularly butyrate produced by Firmicutes species including *Faecalibacterium prausnitzii* and *Akkermansia muciniphila*, are the primary energy source for colonocytes and are essential for maintaining tight junction integrity and regulatory T-cell function. Treg cells, which suppress autoimmune reactivity, are butyrate-dependent. Dysbiosis reduces SCFA production, which simultaneously impairs gut barrier function and reduces the Treg activity that would normally suppress the autoimmune response. This makes microbiome restoration a distinct intervention from gut barrier repair, addressing a different layer of the same problem.

One specific organism within the dysbiosis picture warrants individual attention: *Blastocystis hominis*, a parasitic protozoan found at significantly elevated rates in Hashimoto's patients compared to healthy controls in multiple studies. *Blastocystis* disrupts tight junction proteins, elevates zonulin, and generates antigen traffic through the gut barrier that sustains autoimmune activation. It does not produce prominent gastrointestinal symptoms in most carriers, making it easy to miss without targeted testing. *Blastocystis* is detected on the Gut Zoomer Complete, so no additional panel is required. Its presence should be treated as a contributing driver of gut permeability rather than incidental colonization in the context of active Hashimoto's.

The Gut Zoomer Complete is the appropriate assessment for dysbiosis characterization in Hashimoto's patients. It provides microbiome composition, short-chain fatty acid status, pathogen detection including *H. pylori*, *Blastocystis*, and other parasitic organisms, permeability markers, and secretory IgA in a single panel. The combination of dysbiosis profiling with permeability assessment in a single draw makes it the most clinically efficient gut workup for the Hashimoto's patient who is not responding to initial dietary intervention.

Pathway Six: Epstein-Barr Virus Reactivation

Epstein-Barr Virus (EBV) is the most significant viral trigger in Hashimoto's thyroiditis and the most consistently underinvestigated in clinical practice. Over 90 percent of adults carry EBV as a latent infection following primary exposure, typically in adolescence. The distinction that matters for Hashimoto's is reactivation rather than past exposure: the shift from latent viral dormancy to active viral replication, which generates the immune activation that drives thyroid autoimmunity through multiple overlapping mechanisms.⁸

The molecular mimicry evidence for EBV is more experimentally grounded than for most bacterial triggers. The EBV nuclear antigen EBNA-1 shares structural epitopes with thyroid autoantigens, and cross-reactive antibodies against EBNA-1 and TPO have been demonstrated in patient populations. EBV can also directly infect thyroid follicular cells, establishing a reservoir of viral replication within the thyroid gland itself that generates ongoing immune activation at the site of autoimmune attack. This direct thyroid tropism distinguishes EBV from most other infectious triggers, which operate through systemic immune mechanisms rather than local thyroid infection.

EBV reactivation drives the same Th17 polarization pathway shared by Lyme disease and active Hashimoto's. EBV-driven immune activation suppresses regulatory T-cell function, promotes IL-17 production, and creates the immunological environment in which thyroid autoimmunity escalates. In patients with both EBV reactivation and elevated thyroid antibodies, the viral reactivation is maintaining the immune polarization that makes antibody reduction difficult regardless of dietary and gut interventions.

Testing distinguishes four clinically distinct states. A positive VCA IgM indicates acute or recent primary infection. When VCA IgG is positive but EBNA IgG is negative, the picture is recent primary infection with incomplete seroconversion. Active reactivation, the state most clinically relevant to Hashimoto's, shows as VCA IgG, EBNA IgG, and early antigen (EA) IgG all positive. Past resolved infection with no current reactivation looks the same, except that EA is negative. The full four-marker panel is available through LabCorp and is one of the less expensive items on the testing list. EBV testing is worth considering for any Hashimoto's patient with prominent fatigue, lymph node involvement history, a history of mononucleosis, or a picture that has not moved after the gut, dietary, and bacterial workup is done.

Now the part that most content on this subject leaves out. Everything above is mechanistic and associational. A 2025 review integrates the immunological and virological evidence for EBV across thyroid disease including Hashimoto's, and what it is not is a trial showing that treating a virus changes thyroid antibodies or changes how a patient feels. **There is no validated antiviral protocol for Hashimoto's, and anyone selling you one is ahead of the evidence.** That includes me if I ever describe one to you as established.

So here is what I actually do with a confirmed reactivation pattern, stated at its real evidentiary weight. Prescription antivirals are a drug class that some physicians use for specific viral indications. They are not something I prescribe, and they are mentioned here so you know they exist and can raise them with the physician managing that part of your care. On my side, I support the immune system generally rather than

claiming to treat the virus: sleep, load reduction, correcting documented nutrient deficiencies, and the general immune support tools discussed in Key 8. Those are cheap, low-risk, and defensible on their own terms whatever the virus is doing. What I do not do is build an expensive open-ended antiviral supplement protocol around a marker, because the evidence for that does not exist and the cost is real.

Other Infectious Considerations

Several additional infectious agents have documented or proposed associations with Hashimoto's that are clinically relevant in specific patient contexts.

Hepatitis C Virus: well-established bidirectional relationship. HCV can trigger thyroid autoimmunity through direct thyroid infection and molecular mimicry. Interferon-alpha treatment for HCV is a known thyroid autoimmunity trigger. Relevant for any patient with HCV history or prior interferon treatment. Standard HCV antibody testing applies.

Post-COVID Thyroiditis: SARS-CoV-2 infects thyroid follicular cells via ACE2 receptors, confirmed in autopsy studies.⁹ Post-COVID Hashimoto's-pattern thyroiditis is increasingly documented. For any patient whose antibody escalation followed COVID infection, this context affects treatment framing. No specific additional test; antibody trajectory tracking applies.

Klebsiella pneumoniae and Mycoplasma pneumoniae: cross-sectional association data exist for both. Klebsiella shares thyroid antigen homology through molecular mimicry. Mycoplasma produces immune activation via cold agglutinin mechanisms. Consider in patients with respiratory illness history preceding Hashimoto's onset. LabCorp Mycoplasma IgG/IgM available if clinically indicated.

Parvovirus B19: associated with multiple autoimmune conditions including thyroid autoimmunity. IgG/IgM available through LabCorp. Consider in patients with joint symptoms alongside thyroid antibody elevation.

Lyme Disease and the Thyroid

The biological overlap between Lyme disease and Hashimoto's thyroiditis is clinically important for a specific subset of patients: those in tick-endemic areas whose antibodies are not responding to gut and dietary intervention. The immunological mechanisms are plausible and the shared Th17 pathway is well-characterized. This section presents that evidence accurately, including where it is strong and where it remains at the level of working hypothesis.

The Borrelia-Thyroid Shared Immune Environment

A working hypothesis in functional immunology proposes that *Borrelia burgdorferi* outer surface proteins, particularly OspA and OspC, may share structural similarity with human thyroid tissue antigens. If confirmed, antibodies generated against these proteins during *Borrelia* infection could cross-react with thyroid proteins through molecular mimicry. This hypothesis is currently based on in silico sequence homology analysis and has not been confirmed through experimental demonstration of cross-reactive antibodies in patients or through animal models of Lyme-induced thyroiditis. It should be understood as a plausible but unvalidated mechanism, not as established molecular biology.

What is well-established is that chronic Lyme infection and Hashimoto's share a common immune polarization. *Borrelia* drives a sustained Th17-dominant immune environment, characterized by elevated IL-17, IL-21, and IL-6, with suppressed regulatory T-cell activity. This is the same immune environment that drives lymphocytic thyroid infiltration in Hashimoto's. The two conditions share an immune polarization pattern, which means that active Lyme disease creates and maintains conditions that are broadly permissive to autoimmune thyroid activity, regardless of whether direct molecular mimicry is operating.

The practical clinical implication is that in patients with concurrent Lyme and Hashimoto's, reducing the Lyme burden reduces the Th17 inflammatory environment that sustains the thyroid autoimmunity. This is a mechanistically coherent therapeutic target. The limitation is that no study has directly demonstrated TPO antibody reduction specifically attributable to Lyme treatment, so the claim should be framed as a rational clinical inference from shared immunology rather than a proven treatment outcome.

Symptom Overlap and Missed Diagnoses

Lyme disease and Hashimoto's share a significant symptom overlap. Fatigue, brain fog, cold intolerance, weight changes, hair thinning, anxiety, cognitive impairment, mood instability, and joint pain are features of both conditions. In patients presenting with this symptom cluster, each condition can mask the other. A provider who finds the Hashimoto's stops investigating and misses the Lyme, and one who finds the Lyme misses the Hashimoto's the same way. Patients end up being treated for half of their clinical picture.

The gut represents a third layer of interaction. *Borrelia* infection suppresses secretory IgA production, the gut's primary immune defense against pathogenic colonization. This suppression allows dysbiotic organisms to establish in the gut, promoting intestinal permeability that contributes to autoimmune thyroid activation through the mechanisms described in Key 3. Gut repair in Lyme-Hashimoto's patients is not optional because the Lyme infection is actively degrading the gut barrier that needs to be restored for the thyroid autoimmunity to quiet.

Who Should Be Tested

Tick-borne co-infection testing is indicated for Hashimoto's patients who meet any of the following criteria: have lived in or spent significant time in tick-endemic areas, have a history of a tick bite regardless of whether a bull's-eye rash was observed, have antibodies that are not responding to dietary intervention, gut repair, and *H. pylori* and *Yersinia* workup, have significant neurological or psychiatric symptoms alongside thyroid symptoms, or have been previously treated for Lyme disease but continue to have symptoms.

The Vibrant Tickborne 2.0 PCR panel tests *Borrelia burgdorferi* across 41 antigens, more than the standard Western Blot, and it also covers *Babesia*, *Bartonella*, *Ehrlichia*, and *Anaplasma*, co-infections that most standard panels miss. It is available through rootcauseunlocked.com/order-labs. It is the most expensive panel discussed in this book, which is why it belongs as a secondary investigation after the cheaper gut and infectious workup is done.

Drainage First

Every treatment intervention for Hashimoto's, whether dietary, supplemental, or targeted at an infectious trigger, mobilizes waste products, inflammatory mediators, and cellular debris that must be processed and eliminated through the body's drainage systems. The lymphatic system, the liver, and the kidneys bear the primary burden of this clearance. When drainage capacity is insufficient for the load being generated, mobilized toxins recirculate, inflammatory burden increases, and the autoimmune process can actually flare rather than improve.

This is a leading explanation for why patients who begin aggressive Hashimoto's protocols, particularly antimicrobial protocols for *H. pylori*, *Yersinia*, or Lyme, experience a significant worsening of symptoms in the first weeks. The antimicrobial intervention is working. But the drainage capacity cannot handle the microbial debris load. The result is a Herxheimer reaction that the patient interprets as the treatment making them worse.

At Root Health, I apply the drainage-first sequence to every Hashimoto's case regardless of severity or protocol complexity. The drainage triad runs for a minimum of two weeks before any targeted intervention is introduced, and it is maintained throughout the full protocol.

The Drainage Triad for Hashimoto's

Product	Clinical Role	Dosing
Lymph-Tone II (Energetix)	Extracellular matrix and lymphatic drainage for ECM congestion and xenobiotic accumulation. The appropriate selection for most Hashimoto's patients with dietary toxin burden and compensation history. Taken as drops in warm water.	Per label direction
Methylated B12, liposomal or sublingual	Methylation support for Phase 2 liver detoxification. Low methylation capacity impairs clearance of endotoxins, hormones, and the inflammatory byproducts generated by autoimmune activity.	Per label, twice daily
ReHydration (Energetix)	Electrolyte and mineral foundation for cellular hydration. Required for efficient intracellular waste transport into the lymphatic system.	Per label direction

All three are available through my Fullscript dispensary, wherever you are located.

For patients with significant chronicity, greater than three years of active symptoms, or confirmed Lyme co-infection, a longer drainage phase of three to four weeks before introducing antimicrobial support produces

substantially better clinical outcomes. The investment of time in drainage preparation pays dividends in the form of protocols that are tolerated rather than abandoned.

Lymph Formula Selection

Lymph-Tone I fits acute inflammation, a recent exposure, or an active infection phase.

Most Hashimoto's cases belong on Lymph-Tone II, the formula for extracellular matrix congestion and xenobiotic accumulation.

Lymph-Tone III is for longstanding degeneration, heavy metal burden, or symptoms present more than 3 years.

Never use more than one Lymph-Tone formula simultaneously. All are taken as drops in warm water.

The Environmental Drivers, and Which Ones Actually Hold Up

If you have spent any time reading about Hashimoto's online, you have met the halogen argument. It goes like this. Fluoride, bromide, and chlorine sit near iodine on the periodic table, they are chemically similar enough to take iodine's place, and modern life soaks you in all three. Filter your water, throw out your toothpaste, and your thyroid recovers.

Parts of that are true. The parts that are true are not the parts usually emphasized, and the difference matters, because the version circulating sends people after their toothpaste while the exposure with the best evidence behind it goes unmentioned. Here is what I can defend.

A real mechanism, with a name

Your thyroid pulls iodide out of the bloodstream using a transporter called the sodium-iodide symporter, usually written NIS. It is the front door for every iodine atom your thyroid will ever use. Several environmental anions are shaped enough like iodide to be carried through that same door, and while they occupy it, iodide is not moving.¹³

This is established transporter biology rather than a fringe idea, and medicine once used the same mechanism deliberately: potassium perchlorate was given as a drug precisely because it shuts down thyroidal iodide uptake.

What the mechanism does not tell you is whether the amounts in your actual life are enough to matter. That is a separate question, and it has to be answered compound by compound.

Perchlorate, the one with the strongest case

Perchlorate is the most potent NIS inhibitor of the group, and the thyroid transporter has a higher affinity for it than for iodide itself.¹³ It reaches people through contaminated drinking water and through produce grown with perchlorate-containing irrigation water and fertilizer.

A 2025 systematic review and meta-analysis pooled the human studies on NIS inhibitors and found perchlorate exposure positively associated with TSH across the literature.¹⁴ The effect size is small at population level. It is also the only halogen-type exposure in this chapter with pooled human evidence pointing consistently in one direction. This is the exposure I would prioritize, and it is the one almost nobody talks about.

Bromide, where the mechanism is confirmed and the dose is the question

Bromide does compete at NIS. That is not in dispute. The European Food Safety Authority's 2025 assessment of bromide in food and feed identifies the thyroid as a target organ and states that bromide competes with NIS-mediated iodide uptake.¹⁵

The dose is where the popular version overreaches. In the human data that assessment reviewed, thyroid hormone concentrations shifted at intakes around 9 mg per kilogram of body weight per day, and even at that level the values stayed inside normal limits.¹⁵ The tolerable daily intake the committee derived is 0.4 mg per kilogram per day, and it was set conservatively, with a hundredfold uncertainty factor, to protect neurodevelopment rather than because ordinary diets were producing thyroid disease.

So: a confirmed mechanism, at doses well above what a normal diet delivers. Worth reducing where it is easy. Not the thing that gave you Hashimoto's.

The bromide sources worth knowing are brominated flame retardants in furniture and electronics dust, methyl bromide fumigation residues on some conventionally grown produce, and, until the FDA revoked its authorization in 2024, brominated vegetable oil in citrus-flavored soft drinks. On the flame retardants specifically, a 2024 review of the human literature found the results genuinely heterogeneous, with studies split between positive, negative, and null associations with TSH and T4.¹⁶ I am not going to tell you that finding is stronger than it is.

Fluoride, where I part company with most of what you have read

This is the one where being honest costs me the better story.

The best available dose-response meta-analysis, published in 2024, found that TSH did not change across the lower range of water fluoride exposure and began to rise only above roughly 2.5 mg/L.¹⁷ Community water fluoridation in the United States targets 0.7 mg/L, which sits well below the point where that curve turns. The studies driving the finding were also mostly cross-sectional, mostly conducted in populations with naturally high groundwater fluoride, and mostly in children.

A separate 2024 systematic review looked at the same question, found seven studies that met inclusion criteria, and rated overall confidence in the evidence as low for every outcome, with methodological problems and risk of bias in all seven.¹⁸

Put those together and the evidence position is this: fluoride at the levels found in some natural groundwater does appear to affect thyroid function, and at the levels in American municipal tap water it has not been shown to.

My own position is not the same as where that evidence stops, and I would rather tell you that plainly than let you work it out later. I think limiting fluoride, chlorine, and bromine matters for thyroid health and for general health, and I say so to patients. That is a clinical judgment about total halogen load, and it is not something I can hand you a trial for. Weigh it as what it is. What the published literature currently supports and what I

would do in your position are two different statements, and you are getting both rather than one dressed up as the other.

Practically: if you are on well water, test it, because what is in it is knowable and cheap to find out. On treated municipal water, filtration is still worth doing, and the action list at the end of this chapter says how far I would take it.

Chlorine, where the evidence is absent and I reduce it anyway

Chlorine is the weakest link in the periodic-table argument. I could not find human evidence tying chlorinated drinking water at treatment concentrations to thyroid dysfunction or to thyroid autoimmunity, and I looked. The claim travels because it completes a tidy pattern, not because someone measured it.

So I will not tell you there is a study behind this one. There is not. I still reduce chlorine exposure, for the same reason given in the fluoride section: it belongs to the total halogen load, the reduction is cheap and carries no downside, and a carbon filter that costs very little handles it. That is my judgment rather than a finding, and now you can tell the difference between the two, which is more than most of what you will read on this subject offers you.

The part the halogen story gets backwards

Every halogen argument rests on the same premise: these compounds cause a relative iodine shortage, therefore your thyroid is starved, therefore you should be worried about getting more iodine. In an iodine-deficient population that premise holds. The review literature is consistent that the harms from NIS inhibitors concentrate in iodine-deficient people, in pregnancy, and in children.¹³

But Hashimoto's is not an iodine-deficiency disease, so the premise does not transfer cleanly, and an elevated halogen reading is not by itself a reason to start taking iodine.

I want to be careful here, because my position on iodine is not the one you will find in most Hashimoto's material. I do use iodine clinically in this condition, including kelp, and the diet guide in this kit lays out the reasoning in full: the antibody rise that iodine can produce appears from population data to be a phase rather than a permanent state, thyroid function is minimally disrupted through it, and selenium is the other half of the reaction and goes alongside.²¹²² What I do not do is start iodine because someone's perchlorate came back high. Halogen exposure is an argument for reducing the exposure. Iodine is a separate decision, made deliberately, with selenium, and with monitoring.

Beyond the halogens

Two other exposures have better human data than chlorine and get a fraction of the attention.

Mercury. In an analysis of US women in the 2007 to 2008 NHANES survey, women in the highest blood mercury group had 2.24 times the odds of thyroglobulin antibody positivity, with a confidence interval of 1.22

to 4.12.¹⁹ The same relationship did not appear for thyroid peroxidase antibodies. This is a cross-sectional association in one survey, not a demonstration that mercury causes thyroid autoimmunity, and I want to be precise about that. It is still a finding about thyroid autoantibodies, which is more directly relevant to Hashimoto's than most of the halogen literature.

Bisphenols. A 2025 systematic review and meta-analysis of bisphenol exposure and the hypothalamic-pituitary-thyroid axis found associations with TSH and free T4 in adults, and reported that adult women were more affected than adult men.²⁰ The effect sizes are small and the direction differs by sex, so this is a signal rather than a settled story.

What you are buying in this chapter, stated plainly

Most of this is mechanistic and observational. There is no randomized trial in which reducing an environmental exposure improved Hashimoto's, and I do not expect one, because nobody funds trials on interventions that cannot be patented. What exists is transporter biology that is well characterized, human association studies of varying quality, and regulatory assessments that took the mechanism seriously enough to set intake limits.

I treat that as reason to reduce exposure where reduction is cheap and low-risk, not as reason to reorganize your life.

What I actually do

I do order an environmental exposure panel in Hashimoto's, and I want to be precise about what it does and does not give you.

I use the Vibrant Wellness Environmental Toxins panel, which measures 38 environmental chemicals and includes perchlorate. A urinary panel documents recent exposure. It is suggestive rather than diagnostic for a disease process, and no exposure panel can tell you that a chemical caused your thyroid autoimmunity. Nobody has run a trial showing that testing changes outcomes in Hashimoto's, and I am not going to imply one exists. I order it anyway, because knowing whether perchlorate is actually elevated in you changes how hard I push on water filtration and how closely I watch your iodine status, and because a number in front of someone moves them to act in a way that general advice does not. Read that as my clinical practice rather than as guideline-concordant care.

One limit worth knowing before you pay for it: the panel covers perchlorate, and it does not measure bromide or fluoride. For those two, source control is the only strategy, because there is no test in this workup that will find them for you. The lab guide in this kit covers how I read the panel when it comes back.

Alongside the panel, in this order:

Get iodine status right before anything else. Not by reflex in either direction. Iodine is the most contested question in this condition and the diet guide works through it properly, including where I differ from the standard advice. What matters here is that iodine is decided on its own merits, with selenium alongside and

thyroid function monitored, rather than raised because a halogen result frightened you or dropped because a headline did.

Filter your drinking water. This is the item where I am more aggressive than the evidence strictly requires, and the fluoride section says why. Reverse osmosis is what I suggest when the budget allows, because it removes perchlorate and handles fluoride in the same step. A carbon filter costs far less, reduces chlorine, and is a real improvement over nothing, so it is the answer when reverse osmosis is not realistic. If you are on well water, test it first. Filtration equipment is available from any reputable retailer, and it is not something I sell or profit from.

Reduce the easy bromide sources. Skip the citrus soft drinks, wash and peel conventionally grown produce where practical, and vacuum and dust more often if you have older upholstered furniture, since flame retardants concentrate in household dust.

Handle food storage. Glass rather than plastic for hot food and for anything acidic, and never heat food in plastic. This is the practical end of the bisphenol literature and it costs nothing.

Bring mercury up if your history warrants it. Frequent high-mercury fish consumption, or an occupational exposure, is worth raising with the clinician managing your care. Chelation is not something I do, and not something I would recommend on the strength of the evidence above.

Then let the drainage work from Key 5 carry the load. Reducing what comes in and supporting the systems that clear what is already there are the same project approached from two ends, and the drainage protocol is the half with the clearer clinical rationale.

If you take one thing from this chapter, make it the ranking. Perchlorate is the exposure with the real evidence behind it and the one nobody names, and filtering your water is the single action that addresses it along with the chlorine and fluoride I reduce on judgment rather than on trial data.

The Anti-Inflammatory Diet for Hashimoto's

Diet in Hashimoto's is a targeted clinical intervention, one that addresses specific mechanisms driving the autoimmune process, rather than a general health recommendation. The dietary changes most consistently associated with measurable antibody reduction in clinical research are not general anti-inflammatory eating patterns, though those matter. They are specific to the molecular interactions between dietary components and the immune and gut pathways that sustain the autoimmune attack.

The Removals, in the order I would actually do them

Gluten, handled the way Key 3 describes rather than the way the internet describes. Gliadin is proposed to trigger zonulin release and to generate cross-reactive antibodies against thyroid tissue through molecular mimicry. Both are coherent mechanisms and neither is settled as a driver of your disease. **Test for celiac first, while still eating gluten.** If that is positive, strict elimination is the treatment and strict means labels on every packaged product, asking about preparation when eating out, and treating shared cooking surfaces as a real exposure. If it is negative, an eight to twelve week trial judged on how you feel is reasonable, and a permanent restriction adopted on faith is not.

A note on what the diet evidence actually showed. In the one head-to-head comparison, a Mediterranean pattern outperformed a gluten-free diet on oxidative stress markers, while the gluten-free arm did not move them. No diet in that comparison changed thyroid function or thyroid antibodies. The diet guide in this kit covers it properly. I still think diet matters here. I just think the Mediterranean pattern is the better default and the celiac test is the better first move.

The second removal worth making is conventional dairy, specifically the casein A1 protein present in most commercially available cow's milk products. A1 casein is cleaved during digestion into beta-casomorphin-7, a peptide with opioid receptor activity that has been shown to impair gut barrier integrity and promote intestinal inflammation.¹⁰ In patients with Hashimoto's who are already managing a compromised gut barrier, continuing to consume A1 dairy maintains one of the upstream barrier-disrupting inputs that allows ongoing antigen traffic into systemic circulation. Some patients tolerate A2 dairy, from certain cow breeds, goats, and sheep, without the same permeability consequences.

Refined sugar and high-glycemic carbohydrates create an intestinal environment that favors the dysbiotic organisms most likely to maintain gut permeability and sustain the LPS-driven inflammatory signaling that activates the immune response. For Hashimoto's patients with concurrent *H. pylori* or *Yersinia* colonization, high sugar intake provides a substrate that supports bacterial survival in the gut ecosystem. Removal of refined carbohydrates is both a gut barrier intervention and an anti-pathogenic dietary strategy.

Seed oils, particularly canola, soybean, corn, sunflower, and safflower oils, promote the arachidonic acid cascade and generate pro-inflammatory prostaglandins that amplify the cytokine environment in which

lymphocytic thyroid infiltration occurs. The omega-6 to omega-3 ratio in a standard Western diet is approximately 15 to 20:1. The ratio associated with low autoimmune reactivity is closer to 4:1. Replacing seed oils with olive oil, avocado oil, coconut oil, and fat from grass-fed animal sources shifts this ratio and reduces the inflammatory signaling that drives autoimmune activity at the thyroid.

What to Prioritize

Wild-caught fatty fish, particularly salmon, mackerel, and sardines, provide EPA and DHA that directly resolve inflammatory signaling through specialized pro-resolving lipid mediators. These are not merely anti-inflammatory in a general sense. They actively promote immune resolution. Consuming them three to four times per week during the active treatment phase provides a consistent substrate for inflammatory resolution that cannot be replicated with supplementation alone at the same clinical impact.

Selenium-rich foods deserve particular attention in Hashimoto's. Selenium is essential for the selenoprotein thioredoxin reductase, which regulates the redox environment of the thyroid gland and protects thyroid cells from the hydrogen peroxide generated during hormone synthesis. Brazil nuts are by far the most concentrated food source, with a single nut providing approximately 70 to 90 micrograms. Two to three Brazil nuts per day provides therapeutic selenium intake for most patients. The evidence for selenium in reducing TPO antibody levels is among the more consistent in Hashimoto's nutritional research, though the magnitude of antibody reduction varies across trials.¹¹

Cruciferous vegetables, including broccoli, cauliflower, Brussels sprouts, and cabbage, require individualized guidance in Hashimoto's. Raw cruciferous vegetables contain glucosinolates that can be converted to goitrogens, compounds that interfere with iodine uptake by the thyroid. For patients with adequate iodine status, this is not a significant concern in moderate amounts. Cooking deactivates the majority of the goitrogenic compounds and makes cruciferous vegetables safe for most Hashimoto's patients in standard dietary quantities.

Strategic Supplementation for Hashimoto's

The supplement protocol for Hashimoto's follows the same sequencing principle as every protocol at Root Health: drainage must be established before targeted interventions are introduced, and targeted interventions are selected based on lab findings rather than applied universally. The products named here are the practitioner-grade formulations I use in clinical practice. The framework below explains the clinical reasoning so that you understand what each phase is accomplishing and why the sequence matters.

Phase One: Drainage (Weeks 1 through 2, continued throughout)

A lymphatic botanical, methylated B12, and an electrolyte and trace mineral base form the drainage foundation as described in Key 5. No targeted thyroid or immune interventions are introduced until the drainage triad has been running for a minimum of two weeks. For patients with significant Lyme co-infection or heavy metal burden, three to four weeks of drainage preparation is the clinical standard.

Phase Two: Gut Barrier Repair (Weeks 3 through 6)

Gut barrier repair is the second priority because restoring the physical barrier that prevents ongoing antigen traffic into systemic circulation is prerequisite to meaningful autoimmune modulation. L-glutamine, ideally in a powder that also supplies glycine and proline, provides the amino acid substrate for tight junction protein synthesis and enterocyte regeneration. Bovine colostrum supports the gut-associated lymphoid tissue function that regulates the immune response to gut microbial inhabitants. Bioavailable curcumin provides anti-inflammatory support for the gut environment that accompanies most Hashimoto's cases.

For patients with confirmed celiac disease, strict gluten elimination runs concurrently with gut barrier repair from day one, because there the removal is the treatment. For patients with reactivity on the Wheat Zoomer but negative celiac serology, I run the gluten trial as a time-limited experiment alongside the barrier work and we decide honestly at the end of it.

Phase Three: Thyroid and Immune Support (Weeks 4 through 8)

A thyroid support formula supplying the nutritional cofactors of thyroid hormone production, chiefly selenium, zinc, iron, tyrosine, and vitamin A, is the primary matrix support here. I prefer it over an iodine-focused formula as the default matrix support, because iodine deserves to be a deliberate decision at a known dose rather than an incidental ingredient in a blend. Iodine has a place in this condition. Where I do use iodine, and I do, it goes in knowingly and with selenium alongside. The diet guide covers that reasoning.

Methylated B12 continues from the drainage phase through the full protocol. Beyond its methylation and drainage role, B12 supports neurological function, mood stability, and energy metabolism, all of which are commonly compromised in Hashimoto's patients due to the metabolic burden of chronic autoimmunity.

Vitamin D3 with K2 as MK-7 addresses the vitamin D deficiency that is nearly universal in Hashimoto's patients. Vitamin D functions as an immune modulator, promoting Treg cell activity that suppresses autoimmune reactivity, and correlates inversely with TPO antibody levels in multiple clinical studies.¹² The target is 40 to 80 ng/mL, the same functional range I use across every program, with ongoing monitoring for calcium metabolism in patients requiring extended high-dose supplementation to reach that level.

Phase Four: Targeted Antimicrobial Support (Weeks 6 through 12, based on lab findings)

When EBV reactivation is present on the four-marker panel, general immune support is introduced after drainage and gut barrier repair are established. To restate what Key 3 said, because it matters more than the ingredient list: there is no validated antiviral protocol for Hashimoto's. Lysine is proposed to compete with arginine in viral replication and monolaurin has antiviral activity in laboratory work; both are laboratory findings rather than demonstrated clinical outcomes in this condition, and they should be read that way. Zinc, vitamin C, and adaptogens are general immune support with a reasonable safety profile. EBV is never eradicated in anyone, and the honest framing is that this phase supports the host rather than treating the virus. Hold it to a defined window and judge it on how you feel, not on a titer.

When *H. pylori* is identified, support is introduced after drainage and gut barrier repair are established. Eradication itself is a prescription antibiotic regimen that some physicians use for confirmed *H. pylori* infection. It is not something I prescribe, and it is included here so you know it exists and can raise it with the physician managing that part of your care. This is the one infectious trigger in the book where eradication has outcome evidence behind it, so it is worth having that conversation rather than substituting botanicals for it. The drainage phase matters here because gastric die-off generates an endotoxin load that has to clear.

When *Yersinia* serology is elevated, berberine has shown activity against *Yersinia enterocolitica* in vitro. In vitro activity is a laboratory finding rather than a demonstrated clinical outcome, and it should be read that way. Alongside it, a broad-spectrum botanical antimicrobial provides general support for the wider dysbiotic environment that typically accompanies *Yersinia* colonization. Both sit at the level of reasoned clinical practice, not proven treatment, and both get an endpoint.

When Lyme co-infection is confirmed, the antimicrobial protocol is substantially more involved and requires individualized clinical management based on the specific organisms identified, their burden levels, and the patient's drainage and detoxification capacity. The complete Lyme antimicrobial protocol is addressed in the companion ebook *Unlock Your Lyme*. The sequencing principle is the same: drainage first, then barrier support, then targeted antimicrobial intervention with drainage maintained throughout.

Phase	Products	Duration	Goal
1. Drainage	Lymphatic botanical blend, methylated B12, electrolyte and trace mineral base	Weeks 1 to 2 minimum; continue throughout	Open lymphatic, liver Phase 2, cellular hydration
2. Gut Barrier	L-glutamine with glycine and proline, colostrum, bioavailable curcumin	Weeks 3 to 6	Restore barrier, reduce antigen traffic, repair GALT
3. Thyroid/Immune	Thyroid cofactor formula, methylated B12 (continuing), vitamin D3 with K2, selenium (food or supplemental)	Weeks 4 to 8	Support hormone conversion, modulate immune response, optimize D3
4. Targeted support, by finding	EBV: general immune support (zinc, vitamin C, adaptogens); no validated antiviral protocol exists. H. pylori: eradication with your prescriber. Yersinia: berberine plus broad-spectrum botanical support, on laboratory activity rather than trial evidence. Lyme: see Unlock Your Lyme.	Weeks 6 to 12, based on lab findings	Address identified infectious drivers with drainage and barrier in place

Where to get these

Every item in this protocol is a single ingredient or a standardized botanical, available from any reputable retailer. Look for independent third-party testing (NSF, USP Verified, or equivalent) and match the form named here rather than the name on the front of the bottle.

If you would rather not evaluate that yourself, I keep a curated dispensary of professional-grade products at Root Health, screened for those standards and organized by the phases above.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. Nothing here was included, excluded, ranked, or dosed based on what it earns, and you can buy any of it anywhere.

Your Next Steps with Root Health

The framework in this guide is the clinical picture that most Hashimoto's patients never receive. It identifies the upstream drivers, provides a testing sequence that captures the full autoimmune picture, and lays out a treatment approach organized by mechanistic priority rather than symptom suppression. What it cannot do is replace individualized clinical care. Your specific antibody pattern, your gut findings, your H. pylori and Yersinia status, your nutrient deficiencies, and your history all determine how this framework applies to your case.

Start with the Right Testing

Start cheap and specific, not broad and expensive. For most people that means celiac serology drawn while still eating gluten, a full thyroid panel including free T3, free T4, reverse T3, and both antibodies, and ferritin, B12, folate, zinc, and vitamin D. Low ferritin in particular is extraordinarily common in this population and is the single most frequent reason someone with a good TSH still feels terrible. Add a LabCorp H. pylori stool antigen or urea breath test, which is inexpensive and the one infectious finding with eradication evidence behind it.

The Vibrant Hormone Zoomer captures the thyroid, adrenal, and sex hormone picture in a single draw if you would rather do it that way. Add the Wheat Zoomer when the gluten question is still open after celiac serology. Yersinia serology comes in when the picture has not moved after the gut and dietary work. The Tickborne 2.0 PCR goes last, and only with a real exposure history, because it is the most expensive item on the list. All panels except H. pylori are available at rootcauseunlocked.com/order-labs.

If you want the working half of this program

This book is the reasoning. The parts you actually work from are separate documents, and they exist: **The Hashimoto's Root Cause Kit** collects the lab interpretation guide, the diet guide and the supplement guide in one place. It is the difference between understanding why a panel matters and having the page that tells you how to read yours when it comes back.

If you would rather have everything together, the Ultimate edition bundles this book with all three. Both are at rootcauseunlocked.com/shop.

Neither is required. Everything in this book stands on its own, and the free conversation below is open to you whether you buy anything or not.

Book a Root Discovery Session

The Root Discovery Session is a free clinical conversation with me, and it is a focused clinical intake rather than a sales call. Your symptom history, prior testing, and current picture are reviewed to identify the most targeted path forward. If you are uncertain which workup is appropriate for your case, or if you want clinical interpretation of prior results before spending money on more testing, this is the right starting point.

Book at rootcauseunlocked.com/discovery, or call the Garner, NC office at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, get the celiac test done while you are still eating gluten, and read the lab guide in this kit before you order anything else.

A Personal Note

If you have been on thyroid medication for years and you still do not feel well, that is information. It means the immune attack is continuing. It means the upstream drivers have not been addressed. It does not mean the medication is wrong or that nothing can be done. It means the right investigation has not happened yet.

That investigation is exactly what Root Health is built for.

Dr. Adam Seay, DC · Root Health, Garner NC · rootcauseunlocked.com

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From Understanding to Doing

Everything up to this point was about understanding what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. **Nothing here is a reason to change a thyroid medication dose on your own.** Levothyroxine has a narrow therapeutic window and self-adjusting it is genuinely risky.

Seek urgent care for chest pain, a resting heart rate persistently over 120, severe confusion or drowsiness, or a rapidly enlarging neck mass or difficulty swallowing or breathing. Thyroid disease has emergencies at both ends and they do not wait.

The drivers behind the antibodies

Unlock Your Hashimoto's works through the things that drive the autoimmune attack rather than only the hormone level. Testing exists to find out which of them are yours, so this guide follows the same order.

Driver	What it means	How you find it
The thyroid itself	How much hormone you are making and converting, and how hard the gland is being pushed	TSH, free T4, free T3, reverse T3
The autoimmune attack	Whether your immune system is targeting the gland, and how hard	TPO and thyroglobulin antibodies
The gut	Barrier integrity and microbial composition, the most studied non-thyroid driver	Celiac serology, stool analysis, permeability markers
The raw materials	Selenium, iron, vitamin D, zinc, B12. You cannot make or convert hormone without them	Blood work, covered below
The load	Chronic stress, poor sleep, and infectious burden acting on an already unstable immune system	History, and a small number of targeted tests

The thing to understand about this list: thyroid hormone replacement addresses the first row. It does nothing to the other four. That is not an argument against taking it, and for most people with Hashimoto's it is genuinely necessary. It is the reason people can be perfectly medicated and still feel unwell, and it is what this kit is built around.

The thing most Hashimoto's material gets wrong

You have almost certainly been told your antibodies matter, and that lowering them is the goal.

Antibody titers are the most-measured, most-discussed, and least outcome-validated number in this condition. Several interventions genuinely lower them. **What nobody has demonstrated is that lowering them makes you feel better or changes where your disease goes.**

Keep monitoring them. Just stop treating them as a scoreboard, and be suspicious of anyone selling you a protocol whose success metric is a falling TPO number.

This guide tells you what each test actually establishes, and flags where the evidence stops.

The core panel

Marker	Unit	Standard lab range	Root Health functional target
TSH	mIU/L	0.4 to 4.0	1.0 to 2.5
Free T4	ng/dL	0.7 to 1.7	0.9 to 1.4
Free T3	pg/mL	2.0 to 4.2	2.8 to 3.6
Reverse T3	ng/dL	0 to 24	0 to 15
TPO antibodies	IU/mL	0 to 35	0 to 9
Thyroglobulin antibodies	IU/mL	up to 20	up to 1
TSI / TRAb	%	0 to 140	0 to 99
Ferritin	ng/mL	25 to 300	50 to 150
Vitamin B12	pg/mL	200 to 1100	500 to 900
Folate (RBC)	ng/mL	400 to 2000	600 to 1500
Zinc (plasma)	ug/dL	60 to 120	70 to 110
Selenium (serum)	ug/L	60 to 160	85 to 140
hs-CRP	mg/L	0 to 1.0	0 to 0.5

These are clinical targets used to guide investigation. **They are not validated diagnostic thresholds**, and a result outside a functional window is not a diagnosis of anything.

What each one is actually for.

TSH is the pituitary's opinion of whether there is enough thyroid hormone, not a direct measure of thyroid hormone. It is the most sensitive single screening test and it is also the one most affected by time of day, recent illness, and biotin supplements. Stop biotin for several days before testing, because it interferes with the assay and can produce results that look like thyroid disease when there is none.

Free T4 and Free T3 measure the hormones themselves. Free T4 is the storage form, Free T3 the active form.

Reverse T3 is worth a specific caution. It is elevated by illness, calorie restriction, and stress, which are exactly the situations where the body deliberately downshifts. A high reverse T3 usually reflects that the body is doing something appropriate in response to a stressor, and the useful question is what the stressor is, not how to lower the number.

TPO and thyroglobulin antibodies establish that the process is autoimmune. See the next section for what they do not do.

Ferritin, B12, folate, zinc, selenium are here because their deficiencies produce fatigue, hair loss, and cognitive fog that get attributed to thyroid disease and then persist after the thyroid is well managed. Low ferritin in particular is extraordinarily common in this population and is the single most frequent reason someone with a perfect TSH still feels terrible.

hs-CRP is a nonspecific screening marker. It says an inflammatory load exists somewhere, not where it is coming from, and it confirms nothing on its own. A single elevated reading after an illness means little; repeat it when you are well.

What antibodies do and do not tell you

What they establish: that your hypothyroidism is autoimmune rather than something else, which is diagnostically useful. Positive antibodies with a normal TSH also predict a higher likelihood of developing hypothyroidism later, which justifies monitoring rather than action.

What they do not establish is how you feel, how fast the disease will progress, or whether a treatment is working.

Two things reliably lower them, and here is the honest read on both.

Vitamin D. A meta-analysis of 12 studies covering 862 people found vitamin D supplementation significantly reduced TPO antibody titers, with a standardized mean difference of 1.084, and thyroglobulin antibodies at 0.996.¹ Those are large effects on the antibody number.

Selenium. An umbrella review of 6 systematic reviews covering 75 randomized trials found TPO antibodies fell significantly at 3 months in levothyroxine-treated patients taking selenium, with a standardized mean difference of 0.53. **Only one of the six reviews was rated high quality.**²

And here is what a Cochrane review found when it asked the harder question. Four randomized trials, 463 participants, all at unclear to high risk of bias. **Change in health-related quality of life was not assessed in a single one of them.** Neither was change in levothyroxine dose requirement. The authors concluded the evidence does not allow confident decision making about selenium in Hashimoto's.³

So: the antibody number moves. Whether the person moves with it has largely not been measured. That is the most important sentence in this guide.

The treatment question that is genuinely unsettled

If your TSH is elevated but your free T4 is normal, that is subclinical hypothyroidism, and whether to treat it is a real controversy rather than a settled question your doctor is getting wrong.

A JAMA systematic review and meta-analysis of 21 publications randomizing 2,192 adults found that thyroid hormone therapy lowered TSH into the normal range compared with placebo, and examined the effect on quality of life and thyroid-related symptoms.⁴

A 2026 systematic review in older adults is titled, in its own words, “no evidence of benefit,” evaluating levothyroxine against placebo or no treatment on health-related quality of life.⁵

But it is not one-sided. A meta-analysis of 13 studies in older patients with subclinical hypothyroidism found levothyroxine significantly reduced total cholesterol, triglycerides, LDL cholesterol, and apolipoprotein B.⁶ So there may be a metabolic argument even where there is not a symptomatic one.

What to do with that. If you have been told your subclinical hypothyroidism does not need treating, your doctor is standing on real evidence, not dismissing you. Feeling no different after treatment is also the expected finding rather than a failure on your part. This is a legitimate judgment call, and it is worth having as a conversation rather than a fight.

Reading your results as a pattern

High TSH, low free T4, positive antibodies. Overt autoimmune hypothyroidism. Treatment is not controversial here.

High TSH, normal free T4, positive antibodies. Subclinical, autoimmune. See the section above. Worth monitoring regardless.

Normal TSH, positive antibodies, feeling unwell. Common, and frustrating. The antibodies confirm autoimmunity but do not explain current symptoms. This is where the ferritin, B12, folate, zinc, and vitamin D rows earn their place, along with sleep and the rule-outs below.

Normal everything, feeling unwell. Thyroid has been investigated. The search should continue, not stop.

Normal TSH on treatment, still symptomatic. The most common scenario in the whole condition. Check ferritin first. Then B12 and vitamin D. Then sleep apnea, which is routinely missed in women and produces exactly this picture.

What else produces this picture

Worth ruling out	Why it mimics
Iron deficiency, with or without anemia	Fatigue, hair loss, cold intolerance, poor exercise tolerance
Vitamin B12 deficiency	Fog, fatigue, nerve symptoms
Vitamin D deficiency	Diffuse aching, fatigue, low mood
Obstructive sleep apnea	Unrefreshing sleep, fatigue, fog. Missed constantly in women
Celiac disease	Coexists with Hashimoto's more often than chance, and is a real diagnosis with real testing
Perimenopause	Overlaps almost completely in the age group where Hashimoto's presents
Depression	Fatigue, cognitive difficulty, sleep disruption
Anemia of any cause	Fatigue, breathlessness on exertion

Celiac deserves a note. It genuinely coexists with autoimmune thyroid disease more often than chance would predict. **Get tested while still eating gluten.** Going gluten free first makes the testing unreliable and costs you a clean answer. The diet guide in this kit covers what a head-to-head trial actually found about gluten-free diets in Hashimoto's, and it is not what most thyroid content will tell you.

What to ask your clinician

- “Can we check ferritin, B12, vitamin D, and zinc alongside the thyroid panel? I want to rule out the things that feel like thyroid.”
- “My TSH is in range but I still feel unwell. What else is worth looking at?”
- “Have I been screened for celiac disease, and do I need to be eating gluten for that test to be valid?”
- “Given my TSH and free T4, what is your view on treating versus monitoring, and what would change your mind?”
- “Should I be screened for sleep apnea?”

Bring: your full result history rather than the latest one, since the trend matters more than any single draw; the time of day you were tested; your full medication and supplement list including biotin; and a symptom timeline.

Testing consistency matters more than people realize. Same lab, same time of day, same relationship to your medication dose. TSH varies through the day, and comparing a morning draw to an afternoon one produces changes that are not real.

When to seek clinical review

Urgently: if you have chest pain, a resting heart rate persistently over 120, severe confusion or drowsiness, a rapidly enlarging neck mass, or difficulty swallowing or breathing.

Promptly: new or rapidly worsening symptoms after a dose change, or symptoms of over-replacement including palpitations, tremor, heat intolerance, and unintended weight loss.

Before going gluten free: if celiac is a question, test first.

If you are pregnant or planning pregnancy: thyroid management changes substantially and requirements usually rise. This needs a clinician, promptly, and it is one of the few areas where the evidence for treating subclinical hypothyroidism is stronger.

Track what changes

TSH is checked every few months. How you actually feel changes week to week, and nobody records it, so the conversation defaults to the number.

Root Tracker: rootcauseunlocked.com/tracker

Free, works anywhere, entries stay on your device. Bring the trend to your next appointment.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

The request list above also works in any office.

Get in touch: rootcauseunlocked.com/contact

The gut, which is the most studied driver after the gland itself

Key 3 of the ebook is the gut-thyroid connection, and this is where the research has moved fastest.

What the evidence shows. A systematic review and meta-analysis found altered gut microbiota composition in autoimmune thyroid disease at the family, genus, and species levels compared with controls (PMID 34867823). A separate study characterizing the microbiome in primary hypothyroidism concluded that hypothyroidism causes changes in the gut microbiome and that, in mice, an altered flora can in turn affect thyroid function, which is what makes it an axis rather than a one-way street (PMID 32519746).

A 2026 review of the gut-thyroid axis describes dysbiosis, impaired intestinal barrier function, and altered microbial metabolites, particularly short-chain fatty acids, as contributors to immune imbalance, and calls the gut microbiota **a modifiable contributor to thyroid autoimmunity** while naming the honest gap: causality still needs clarifying (PMID 41292506).

Read that carefully. “Modifiable contributor” with causality unresolved is a fair reason to work on the gut. It is not a promise that fixing the gut resolves the antibodies.

What to ask for. Celiac serology first, drawn while you are still eating gluten. Then, if gut symptoms are present, a comprehensive stool analysis. The companion Gut Root Cause Kit covers barrier markers in more depth, and the caution there applies here too: read a barrier panel as a pattern, not as proof from any single marker.

The load (stress, sleep, and infection)

This section was missing from the first version of this kit, and it is a genuine part of the picture rather than a wellness postscript.

Stress, with the citation limits stated plainly. The controlled human work linking stressful life events to autoimmune thyroid disease is in **Graves' disease**, not Hashimoto's. One controlled study found an excess of life events in the year before Graves' onset (PMID 8498147), and another found that patients with Graves' reported more negative stressful life events than a comparison group with toxic nodular goitre, suggesting life events act as a precipitating factor (PMID 11453947).

I am telling you the disease those studies were done in, because most sources citing them do not. The honest position is that stress is an established precipitant in one autoimmune thyroid disease and a reasonable clinical concern in the other, not a proven cause of Hashimoto's.

What follows from that. A daily down-regulating practice, protected sleep, and a realistic look at load are worth doing on their own terms. They are free, they carry no interaction risk, and the mechanism they act on, immune regulation, is the same one the antibodies live in.

- Five to ten minutes daily of slow breathing with a longer exhale than inhale, done consistently.
- A fixed sleep window of seven to nine hours. Hypothyroid fatigue and poor sleep hygiene compound each other, and only one of them is treatable with a tablet.
- Movement you can sustain. Not high-intensity work on a bad fatigue day, which reliably backfires.

Infection, as a plausible driver rather than a certainty. Epstein-Barr virus has drawn sustained attention for a potential role across thyroid disease including Hashimoto's, and a 2025 review integrates the immunological and virological evidence (PMID 41229422). Note what that is: a mechanistic and associational review, not a trial showing that treating a virus changes thyroid antibodies. There is no validated antiviral protocol for Hashimoto's, and anyone selling you one is ahead of the evidence.

Environmental exposure, and the one panel I order for it

Halogens come up constantly in thyroid conversations, usually as the claim that fluoride, bromide, and chlorine are crowding iodine out of your gland. Key 6 of the ebook works through what actually holds up in that literature, compound by compound. This section is narrower. It covers what I test for, how I read it, and what the test cannot tell you.

The mechanism, briefly. Your thyroid takes up iodide through the sodium-iodide symporter, usually written NIS. Several environmental anions are close enough in shape to be carried through the same transporter, and perchlorate is the most potent of them, with the transporter showing higher affinity for perchlorate than for iodide itself.¹³ Pooled human data associate perchlorate exposure with higher TSH.¹⁴

What I order. The Vibrant Wellness Environmental Toxins panel, a urinary panel covering 38 environmental chemicals, of which perchlorate is the one most directly tied to thyroid function. I want to be straight with you about its standing: no trial has shown that running this panel improves outcomes in Hashimoto's, and I am not going to suggest otherwise. I order it because an elevated perchlorate result changes how hard I push on water filtration and how closely I watch iodine status, and because a number in front of someone tends to produce action where general advice does not. That is my clinical practice rather than guideline-concordant care, and you are entitled to know which one you are getting.

What it does not measure. Bromide and fluoride are not on this panel. There is no test in this workup that will find them for you, so for those two you are relying on source control alone. If you are on well water, testing the water itself is the useful move, and it is inexpensive.

Reading a perchlorate result

What it tells you	What it does not tell you
That perchlorate was in your system recently, at roughly what level	That perchlorate caused your thyroid autoimmunity
Whether your exposure sits high or low relative to other people	Your average exposure over months, from one sample
Whether water filtration is worth prioritizing for you	Anything about your bromide or fluoride exposure

Two limits are worth understanding before you read your own result.

It documents recent exposure, not accumulated burden. Perchlorate has a short biological half-life and is excreted quickly in urine.¹⁵ A result reflects roughly the last day, not the last year. This is not a body-burden measurement, and anyone who tells you it is has misunderstood the test.

A single sample is a rough category, not a precise number. In a study tracking children over six months, urinary perchlorate showed poor within-person reproducibility, with an intraclass correlation of 0.10 and more variation inside the same child than between different children.¹⁵ The same study found that a single spot sample still ranked people into exposure categories reasonably well. So read your result as high, middling, or low rather than as a figure to track to the decimal, and do not chase small movements between draws.

What I do with an elevated result

The response is source control, and water is the main lever, because filtration is inexpensive, has no downside, and addresses perchlorate along with the chlorine and fluoride the panel never measured. Reverse osmosis is the option that removes perchlorate. Produce grown with contaminated irrigation water is the other common route, with leafy greens the usual example, and washing helps rather than solves it.

I also look harder at iodine status when perchlorate is elevated, and I want to be careful about what that does and does not mean. The reasoning that perchlorate competes with iodide leads people straight to the conclusion that they should take iodine to win the competition. My position on iodine in Hashimoto's is more permissive than most thyroid material, and the diet guide sets it out in full with the evidence on both sides. But a perchlorate result is not the thing that should trigger that decision. Reduce the exposure, which is what the result is actually telling you to do. If iodine is going to be part of your plan, it gets decided on its own merits, at a known dose, with selenium alongside, and with your thyroid function monitored, and that is a conversation rather than a reaction to one line on a panel.

A normal result does not close the question. Given the short half-life, a low reading tells you that your exposure in the last day was low. If you drink unfiltered water from a source you have never tested, the sensible move is still to test the water or filter it, whatever the urine said.

Where this sits in the workup

This is not a first-line test. The core thyroid panel, the antibodies, and the nutrient markers earlier in this guide come first, and the gut and load sections cover the drivers with the larger evidence base behind them. I run the environmental panel when the picture is not adding up, when someone has a known exposure history or an untested water source, or when we have worked the higher-yield items and want the next layer. Ordering it first would be spending your money in the wrong order.

What almost nobody checks, and what it changes

Everything above is the standard workup done properly. This section is different. These are four things that routinely go unexamined, each of which can change how your panel reads or how well your treatment works, and none of which shows up in a normal thyroid visit.

Whether your stomach can actually absorb your medication

This is the one I wish more people knew about, because it explains a specific and common frustration: your dose keeps going up and your TSH will not settle.

Levothyroxine needs stomach acid to dissolve before your intestine can absorb it. If acid production is low, the tablet stays in a form your gut cannot take up, and the dose on the bottle is not the dose reaching you. In a study published in the New England Journal of Medicine, the daily thyroxine requirement was 22 to 34 percent higher in patients with H. pylori-related gastritis, atrophic gastritis, or both, compared with a reference group. When H. pylori was eradicated, the effect largely reversed.¹⁶ A broader review of levothyroxine absorption reaches the same place and adds celiac disease, lactose intolerance, and several drug interactions to the list of things that get between the tablet and you.¹⁷

Two reasons this matters more in Hashimoto's than in ordinary hypothyroidism. First, H. pylori shows up in the ebook already, as one of the infectious triggers implicated in thyroid autoimmunity through molecular mimicry. If it is present, it may be doing two jobs at once: contributing to the autoimmune process and blocking the treatment for it. Second, autoimmune atrophic gastritis clusters with autoimmune thyroid disease, so the population most likely to have this problem is exactly the population being treated.

What to do about it. If you are on levothyroxine, taking it correctly, and your TSH still will not come into range, absorption belongs on the list before the dose goes up again. Ask about H. pylori testing and, if you have the symptoms for it, about atrophic gastritis. This is a conversation with the clinician managing your prescription.

Whether your estrogen is moving your numbers

Most of your thyroid hormone travels bound to a carrier protein called thyroxine-binding globulin. Oral estrogen raises the amount of that carrier, which increases the bound fraction and lowers the free, usable fraction. In women being treated for hypothyroidism, oral estrogen therapy can raise the levothyroxine dose requirement for this reason.¹⁸

The practical version: starting or stopping an oral contraceptive, or starting or stopping oral hormone replacement, can shift your thyroid numbers without anything having changed in your thyroid. If your labs moved and nobody asked what else changed that month, this is worth raising. The route matters too, since the effect comes from oral estrogen passing through the liver first.

Whether your gut is doing part of the conversion

Most of the active thyroid hormone in your body, T3, is made outside the thyroid by converting T4. The liver and kidneys do the bulk of that work. The gut appears to contribute as well, which the usual explanation leaves out.

Deiodinase activity has been identified in the intestinal wall and may add to total body T3. Separately, bacterial enzymes in the gut, specifically sulfatases and beta-glucuronidases, regulate the deconjugation of thyroid hormones, which affects how

much gets recycled back into circulation rather than excreted. Some gut bacteria appear to bind thyroid hormones and act as a reservoir.¹⁹

I want to be careful about the weight of this. Much of it comes from animal and laboratory work rather than from human outcome trials, and the findings are not all pointing the same direction. It is a proposed mechanism, not a settled one. What it does is give a concrete reason why the gut chapters in this kit are not a change of subject from your thyroid, and why gut repair sometimes produces thyroid improvement that the antibody number does not explain.

Whether insulin resistance is sitting underneath it

Fasting insulin is not part of a standard thyroid workup and I run it anyway, because insulin rises years before glucose does and a normal glucose tells you nothing about how hard your body is working to keep it there.

The thyroid link specifically is less settled than the general case for measuring it. A population-based cohort of euthyroid individuals examined thyroid hormones, thyroid antibodies, and insulin resistance together, and the literature as a whole is mixed on whether thyroid autoimmunity independently tracks with insulin resistance in people whose thyroid function is normal.²⁰ So I am not going to tell you that insulin resistance drives Hashimoto's.

What I will tell you is that the symptom overlap is near total. Fatigue, weight that will not move, brain fog, and afternoon crashes are the presenting complaint in both, and a great many people arrive convinced their thyroid explains everything when their fasting insulin has never been drawn. Finding it does not mean your thyroid is fine. It means you have found a second thing, and it is one of the more addressable things on this list.

The common thread

Each of these is a case where the thyroid panel is not wrong, it is just incomplete. A number that will not move, a number that moved for a reason nobody asked about, a conversion step happening somewhere nobody looked, and a parallel problem wearing the same symptoms. None of them requires an exotic test. They require someone to ask the question.

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Functional target ranges transcribed from the Root Health blood chemistry range.

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing here is a reason to change a thyroid medication dose on your own.

This guide is going to disagree with the most common piece of Hashimoto's diet advice on the internet. Gluten still gets its due here. What I will show you is what the evidence says, and why the standard advice has the order of operations backwards.

Start here: the medication timing detail

Before any diet change, this one is worth more to most readers than everything else in the guide, and it takes no willpower at all.

Levothyroxine absorption is reduced by things people take at breakfast. The prescribing information for levothyroxine lists interference from calcium supplements, iron supplements, and other minerals, and absorption is affected by food generally. Coffee and soy are also commonly implicated.

What to do. Take it on an empty stomach with water, and separate it from coffee, food, calcium, iron, magnesium, and multivitamins. Most guidance suggests 30 to 60 minutes before eating, and separating minerals by around four hours.

Why this matters more than it sounds. If your absorption is inconsistent, your dose is inconsistent, and no diet change will show through that noise. People spend months on elimination diets while taking their thyroid medication with a mineral-fortified breakfast. Fix the timing first, hold it steady for six weeks, and then judge everything else against a stable baseline.

Do not change your dose to compensate. Change the timing, tell your prescriber you have done so, and retest when they suggest.

The gluten question, and why the usual advice is backwards

The standard recommendation is: you have Hashimoto's, go gluten free. Two separate lines of evidence say that is the wrong first move.

A head-to-head comparison. A pilot study compared the Mediterranean diet, a gluten-free diet, and a free diet in people with Hashimoto's, measuring oxidative stress markers over 12 weeks.¹

The Mediterranean group showed significantly lower advanced glycation end products and significantly higher glutathione peroxidase, thioredoxin, and total antioxidant capacity, both compared with their own baseline and compared with the other two groups. **In the gluten-free group, the oxidative stress parameters did not change significantly.**

And one finding applied to all three arms. **There was no significant change in TSH, free T4, thyroglobulin antibodies, or TPO antibodies in any group.** No diet moved thyroid function or thyroid antibodies in twelve weeks.

The authors' own conclusion is that a gluten-free diet does not significantly influence markers of oxidative stress or thyroid autoimmunity and function. It is a pilot study and should be read as one.

A genetic causality analysis. A Mendelian randomisation study examined whether the relationships here are causal.² Two findings.

Hashimoto's disease **increases the risk of celiac disease**, with an odds ratio of 1.544 and a confidence interval of 1.153 to 2.068. That association holds, and it runs in that direction.

And the study "does not support genetic liability related to GFD with AITD," concluding that **a gluten-free diet has no protective effect on autoimmune thyroid disease** and that any benefit people experience may work through other mechanisms.

A 2026 review states the position well: in celiac disease the pathogenic role of gluten is firmly established, but in Hashimoto's, gluten is better viewed as a **candidate modifier rather than a proven universal trigger**.³

So what should you actually do

Get tested for celiac disease, while still eating gluten. The standard advice skips this step, and it is the one that matters. Hashimoto's raises your celiac risk measurably. Celiac is a real diagnosis with real testing and a treatment that genuinely works, and the testing depends on you currently eating gluten. Go gluten free first and you lose the ability to get a clean answer without a deliberate gluten challenge later.

If celiac testing is positive, gluten free is the treatment. Not a wellness experiment, a treatment, and it matters enormously.

If celiac testing is negative and you still want to try it, that is reasonable. Run it as a time-limited experiment with a defined endpoint, judged on how you feel rather than on a repeat antibody panel, since the antibody panel did not move in the trial. Give it eight to twelve weeks and then decide honestly.

What I would not do is adopt a permanent restriction, on the strength of evidence that does not support it, in a condition where the same trial showed a Mediterranean pattern outperforming it.

The pattern that did outperform

In the only head-to-head in this guide, the Mediterranean pattern was the one that moved markers.¹ It is also less restrictive, cheaper, easier to sustain, and better supported for general health than any elimination diet.

Build meals from:

Vegetables in volume and variety, across colors. Olive oil as the main fat. Fish, particularly oily fish, a couple of times a week. Legumes. Whole grains if you tolerate them and celiac has been excluded. Nuts and seeds. Fruit, with an emphasis on berries. Herbs and spices generously.

Moderate: red meat, and processed meat more so. Refined carbohydrate and added sugar. Alcohol.

This is not a thyroid intervention and I am not going to sell it as one. No diet in that trial changed thyroid function or antibodies. What the Mediterranean arm changed was oxidative stress markers, which is a mechanistic outcome rather than a symptom one. It is a good way to eat with the best available support in this specific condition. That is the honest size of the claim.

Iodine, the most contested question in Hashimoto's

This is the question I get asked most, and it is the one where I differ from a lot of thyroid material, including some of what you will find under my own condition's name elsewhere on the internet.

Iodine is required to make thyroid hormone. There is no thyroid hormone without it. That much is not controversial.

The case against iodine in Hashimoto's has real evidence behind it, and I am not going to hide it from you. In a study of risk factors for autoimmune thyroid disease, median urinary iodine was significantly higher in the autoimmune group than in controls.⁴ Mechanistic work found that excessive iodine activates a hypoxia pathway promoting apoptosis of thyroid follicular cells, described by the authors as potentially an important risk factor contributing to Hashimoto's development.⁵ Those are the findings that produced the standard advice to avoid iodine, and if you have read that advice elsewhere, it did not come from nowhere.

What that advice usually leaves out is the time course. The largest natural experiment on this is what happened when countries introduced salt iodization. A 2024 review tracked the pattern across populations and found antibody prevalence rising in the first several years, then declining, and then settling back toward pre-iodization levels over the longer term. In Denmark, TPO antibody prevalence rose from 14.3 percent to 23.8 percent within about five years. In Sri Lanka, a follow-up group went from 46.9 percent down to 23.8 percent over the next three years, and twenty years on the prevalence sat lower than it had during the peak. Throughout all of it, the authors describe minimal disruption of actual biochemical thyroid function, with overt or subclinical dysfunction remaining uncommon.¹³

So the antibody rise is real. What the population data suggest is that it is a phase rather than a permanent state, and that it is not accompanied by the thyroid failure you would expect if the antibodies told the whole story.

Selenium is the part of this that is not optional. Making thyroid hormone requires hydrogen peroxide, which thyroid peroxidase uses to oxidize iodide. Selenium-dependent glutathione peroxidase is what clears that hydrogen peroxide before it damages the gland, which is why selenium status matters most precisely when iodine intake goes up.¹⁴ If you are going to raise iodine, selenium is not a nice addition, it is the other half of the same reaction. The supplement guide in this kit covers the dose.

Where I land. I use iodine clinically, including kelp, in Hashimoto's patients. What I see matches the pattern above. Antibodies often rise, then come back down, and people feel better in the meantime. That is my clinical observation across the patients I have treated, not a controlled trial, and you should file it accordingly. Dr. David Brownstein has written extensively on iodine and reports the same rise-then-settle pattern from his own practice; his work is published as books and clinical experience rather than randomized trials, and I am telling you that so you can weigh it for what it is.

I hold this position knowing the literature above runs the other way on the association. The reason I hold it is that the antibody number is not the outcome I am treating. Which brings up the thing this guide says elsewhere and will say again here.

Do not judge this on your antibody titer. The lab guide in this kit explains why antibody levels are a poor success metric in general, and this is the clearest example of it. If your antibodies rise on iodine and you feel better, the titer is not the emergency it looks like. Feeling worse while they rise is different: that is information, and a reason to stop and talk to me rather than push through.

How to actually do it. Not alone, and not by working up to a large dose because a book said so. Pair it with selenium from the start. Have your thyroid function rechecked rather than tracking antibodies alone. Kelp in particular delivers large and highly variable doses depending on the product, so if you use it, use a product that states its iodine content. And if you are on

thyroid medication, changing your iodine intake is a conversation with the clinician managing that prescription, because your dose requirement can shift.

If you are pregnant or planning pregnancy, iodine requirements rise and the stakes are higher on both sides. That is a conversation with your clinician, not a decision to make from a guide.

Halogens, and why the usual advice points the wrong way

Right after the iodine conversation, most people run into the halogen argument, which says that fluoride, bromide, and chlorine are chemically similar to iodine, that they crowd iodine out of your thyroid, and that removing them is the fix. It is worth walking through, because the compounds are not equal and the usual ranking of them is wrong.

The mechanism is real. Your thyroid takes up iodide through a transporter called the sodium-iodide symporter. Several environmental compounds are close enough in shape to ride that same transporter, and while they do, iodide is not moving through it.⁸ That is established biology, not a fringe claim.

The compounds are not equal, and the ranking is not the one you have read.

Perchlorate is the most potent of them, and the transporter actually prefers it to iodide.⁸ It shows up in some drinking water and in produce grown with contaminated irrigation water. A 2025 meta-analysis of the human studies found perchlorate exposure positively associated with TSH.⁹ It has the best evidence of the group and gets the least attention.

Bromide does compete at the transporter, confirmed in the European Food Safety Authority's 2025 assessment, which names the thyroid as a target organ.¹⁰ But in the human data reviewed there, thyroid hormone levels only shifted at intakes around 9 mg per kilogram of body weight per day, and even then stayed within normal limits.¹⁰ That is far above what a normal diet delivers. Reduce it where it is easy, which mostly means citrus-flavored soft drinks and, if you want to go further, conventionally grown produce that may carry fumigation residue.

Fluoride is where the published evidence thins out considerably. The best dose-response meta-analysis found no change in TSH across the lower exposure range, with levels beginning to rise only above roughly 2.5 mg/L in water.¹¹ US municipal fluoridation targets 0.7 mg/L. A separate 2024 systematic review rated overall confidence in the fluoride and thyroid evidence as low across every outcome it examined.¹² If you are on well water, testing it is cheap and sensible.

Chlorine completes the pattern on the periodic table and not much else. I could not find human evidence linking chlorinated drinking water at treatment concentrations to thyroid problems, and I looked for it.

Where I land, which is not identical to where the evidence stops. I think limiting fluoride, chlorine, and bromine is worth doing for your thyroid and for your health generally, and I tell patients so. That is my clinical judgment about total halogen load rather than a conclusion I can point you to a trial for, and you should be able to tell those two things apart. Filtering your water is inexpensive, carries no downside, and covers all three at once, so the cost of my being wrong about it is close to zero. That is the basis I act on.

Now the part that matters most for your diet. Every version of the halogen argument concludes that you are effectively iodine-starved, and the intuitive next step is to take more iodine to win the competition. The harms from these compounds are in fact documented mainly in iodine-deficient populations, in pregnancy, and in children.⁸ Hashimoto's is not an iodine-deficiency condition, so that reasoning does not transfer cleanly, and a halogen result is not by itself a reason to start iodine.

The iodine section above covers where I land on iodine itself, which is more permissive than most thyroid material and comes with selenium attached. The point here is narrower: reduce these exposures because reducing them is cheap and sensible, not because a halogen reading tells you to go take something.

What that looks like in practice. Filter your drinking water. Reverse osmosis is the better option where the budget allows, since it removes perchlorate and handles fluoride in one step, and a carbon filter costs much less, reduces chlorine, and beats doing nothing. If you are on well water, test it first. Skip the citrus soft drinks. Use glass rather than plastic for hot and

acidic foods. And treat iodine as its own decision, made the way the iodine section describes, rather than as a reflex response to a halogen worry.

Cruciferous vegetables, which are probably fine

Broccoli, cabbage, kale, and Brussels sprouts contain compounds that can interfere with iodine uptake, which is why they carry a reputation in thyroid circles.

The practical reality is that the amounts required to affect thyroid function are far beyond normal consumption, cooking reduces the effect substantially, and these are among the more nutritious foods available. In a population where iodine excess is the more relevant concern than iodine deficiency, avoiding them is solving the wrong problem.

Eat them. If you are eating raw kale in very large daily quantities and have low iodine intake, that is worth mentioning to your clinician. Otherwise this is a worry you can put down.

The nutrients that actually explain how you feel

The lab guide in this kit covers testing. From a food standpoint:

Iron is the big one. Low ferritin is extremely common in this population and produces fatigue, hair loss, and cold intolerance that get attributed to the thyroid and then persist after the thyroid is managed. Red meat, poultry, fish, legumes, and dark leafy greens, with vitamin C at the same meal to aid absorption from plant sources. **Take iron supplements away from levothyroxine.**

Selenium is found in seafood, eggs, and Brazil nuts. Brazil nuts are extremely concentrated and vary enormously nut to nut, so a couple a day is plenty and a handful is too many.

Zinc from shellfish, meat, legumes, and seeds.

Vitamin D is difficult to get from food in meaningful quantities. Testing beats guessing.

B12 from animal foods, with the caveat from the lab guide that certain medications interfere with absorption.

Running a fair experiment

Fix the medication timing first. hold it six weeks, and let a stable baseline establish itself. Everything else is measured against that.

Then change one thing. Mediterranean pattern first, since it is the one with head-to-head support and it costs you the least.

Give it eight to twelve weeks. The trial ran twelve.

Do not judge it on antibodies. No diet arm in the trial moved them. Judge it on energy, sleep, cognition, and how you actually feel, recorded rather than remembered.

Track it: rootcauseunlocked.com/tracker

Realistic expectations

No diet in the available head-to-head changed thyroid function or thyroid antibodies.¹ A gluten-free diet has no demonstrated protective effect on autoimmune thyroid disease.² Gluten is a candidate modifier, not a proven universal trigger.³

What diet realistically offers here is a better background: lower oxidative stress on a Mediterranean pattern, better nutrient status for the deficiencies that masquerade as thyroid symptoms, and the avoidance of two specific mistakes, excess iodine and poorly timed medication.

That is worth doing. It is not a cure and it will not replace treatment.

And the most valuable single action in this guide remains the celiac test, taken while still eating gluten, because that is the one path where a dietary change is genuinely the treatment.

When to seek clinical review

Urgently: if you have chest pain, a resting heart rate persistently over 120, severe confusion or drowsiness, or difficulty swallowing or breathing.

Before going gluten free: get celiac testing while still eating gluten.

Before taking iodine or kelp: ask, particularly with autoimmune thyroid disease.

If you change medication timing: tell your prescriber, because absorption may improve and your dose may need review.

If pregnant or planning pregnancy: thyroid and iodine requirements both change and this needs clinical management.

The gut connection behind the diet

Diet is one input into the gut-thyroid axis, so it helps to know what that axis looks like. A systematic review and meta-analysis found altered gut microbiota composition in autoimmune thyroid disease compared with controls (PMID 34867823), and a 2026 review describes dysbiosis, impaired barrier function, and altered short-chain fatty acid production as contributors to immune imbalance, calling the microbiota a modifiable contributor while noting that causality is still being worked out (PMID 41292506).

Practically that means fiber diversity and fermented foods have a rationale here beyond general health advice, and it also means no diet has been shown to reverse thyroid autoimmunity. Both things are true.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

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Levothyroxine absorption guidance reflects the product prescribing information rather than a specific trial.

For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

Root Health | Dr. Adam Seay, DC | Garner, NC | rootcauseunlocked.com

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. **Nothing here is a reason to change a thyroid medication dose on your own.**

Every supplement in this guide has to be timed around levothyroxine. It is the most consequential practical point in the document, which is why it gets its own section.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed, with some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions. It names the funding and the evidence tier, and lets you see both.

The load (stress, sleep, and infection)

This section was missing from the first version of this kit, and it is a genuine part of the picture rather than a wellness postscript.

Stress, with the citation limits stated plainly. The controlled human work linking stressful life events to autoimmune thyroid disease is in **Graves' disease**, not Hashimoto's. One controlled study found an excess of life events in the year before Graves' onset (PMID 8498147), and another found that patients with Graves' reported more negative stressful life events than a comparison group with toxic nodular goitre, suggesting life events act as a precipitating factor (PMID 11453947).

I am telling you the disease those studies were done in, because most sources citing them do not. The honest position is that stress is an established precipitant in one autoimmune thyroid disease and a reasonable clinical concern in the other, not a proven cause of Hashimoto's.

What follows from that. A daily down-regulating practice, protected sleep, and a realistic look at load are worth doing on their own terms. They are free, they carry no interaction risk, and the mechanism they act on, immune regulation, is the same one the antibodies live in.

- Five to ten minutes daily of slow breathing with a longer exhale than inhale, done consistently.
- A fixed sleep window of seven to nine hours. Hypothyroid fatigue and poor sleep hygiene compound each other, and only one of them is treatable with a tablet.
- Movement you can sustain. Not high-intensity work on a bad fatigue day, which reliably backfires.

Infection, as a plausible driver rather than a certainty. Epstein-Barr virus has drawn sustained attention for a potential role across thyroid disease including Hashimoto's, and a 2025 review integrates the immunological and virological evidence (PMID 41229422). Note what that is: a mechanistic and associational review, not a trial showing that treating a virus changes thyroid antibodies. There is no validated antiviral protocol for Hashimoto's, and anyone selling you one is ahead of the evidence.

The finding this guide is built around

Two studies, read side by side, tell you something no single one of them says.

The intervention that most improved quality of life did not change antibodies at all. In a double-blind randomised study, women with Hashimoto's received nutritional education plus either a probiotic (*Lactiplantibacillus plantarum* 299v) or placebo. The probiotic group improved overall quality of life far more, 60.94 points versus 35.94, and improved in 12 of 14 domains versus 3 of 14.¹

And: anti-TPO levels did not improve in either group. The authors conclude the benefits “appear to be independent of anti-TPO levels.”

Meanwhile the interventions that reliably lower antibodies have never been shown to make anyone feel better. The Cochrane review of selenium found that health-related quality of life was not assessed in a single one of the four trials.

Put those together and you get the operating principle for this whole guide: **the antibody number and how you feel are two different outcomes, and the things that move one are not necessarily the things that move the other.** Most Hashimoto's supplement advice optimizes the number. This guide tries to optimize you.

Timing, before anything else

Levothyroxine absorption is reduced by minerals and by food. The prescribing information lists interference from calcium and iron supplements in particular.

The rules:

- Levothyroxine on an empty stomach with water, 30 to 60 minutes before eating
- **Calcium, iron, magnesium, zinc, and multivitamins separated by around four hours**
- Coffee separated from the dose
- Consistency matters more than perfection. Same routine daily beats an optimal routine done erratically

Why this outranks every supplement below. If your absorption swings, your effective dose swings, and nothing you add will be interpretable through that noise. Fix timing, hold it six weeks, then start experimenting.

If you change your timing, tell your prescriber. Absorption may improve, which can mean you are now on a higher effective dose than intended.

Tier 1: The one with the most support

Selenium

What the evidence shows, in both directions.

A network meta-analysis comparing different supplements in Hashimoto's concluded, after examining the alternatives, that patients should be given an appropriate amount of selenium as an auxiliary treatment alongside standard care.² An umbrella review of 6 systematic reviews covering 75 trials found TPO antibodies fell significantly at 3 months in levothyroxine-treated patients on selenium, standardized mean difference 0.53, though only one of the six reviews rated high quality.³

And the Cochrane review, which asked the harder question. Four trials, 463 participants, all at unclear to high risk of bias. **Change in quality of life was not measured in any of them.** Neither was change in levothyroxine dose requirement. The authors concluded the evidence does not permit confident decision making.⁴

My honest read. Selenium has the best antibody evidence of anything here and the weakest evidence that it changes your life. It is reasonable to trial, it is inexpensive, and you should not expect to feel it.

Commonly studied amount: 200 mcg daily in the trials. **Functional serum target I use at Root Health:** 85 to 140 ug/L. This is one worth testing rather than assuming. **Cautions, and these matter:** selenium has a genuinely narrow window. Chronic high intake causes selenosis, with hair and nail changes, garlic breath, and gastrointestinal upset. **Do not stack multiple products containing selenium**, and remember Brazil nuts are extremely concentrated and vary enormously nut to nut. A couple a day is plenty; a handful is too many. This is not a more-is-better nutrient.

Tier 2: Depends on your situation

Iron, if your ferritin is low

Not a thyroid treatment. It is on this list because **low ferritin is the single most common reason someone with a well-controlled TSH still feels terrible**, and because the symptoms overlap almost completely (fatigue, hair loss, cold intolerance, poor exercise tolerance).

Functional ferritin target I use at Root Health: 50 to 150 ng/mL, against a standard range of 25 to 300. A great many people sit between 25 and 50, get told it is normal, and are symptomatic.

Cautions: test before supplementing, because iron overload is a real condition and iron is not a benign supplement.

Separate from levothyroxine by four hours. Take with vitamin C, away from calcium and coffee.

Vitamin D, where two meta-analyses disagree

I am going to show you the disagreement rather than pick the answer I prefer.

One meta-analysis of 12 studies covering 862 people found vitamin D significantly reduced TPO antibodies, standardized mean difference 1.084 with a confidence interval of 1.624 to 0.545, and thyroglobulin antibodies at 0.996.⁵

A network meta-analysis found vitamin D did not effectively reduce TPO antibodies, with a standardized mean difference of 2.54 and a confidence interval running from 6.51 to positive 1.42, which crosses zero.²

Different methods, different included trials, opposite conclusions. When two meta-analyses of the same question disagree this cleanly, the honest summary is that the answer is not settled.

What I would do anyway: correct a documented deficiency, because that is defensible on general grounds regardless of what it does to antibodies. Do not take it as an antibody treatment.

Functional target: 40 to 80 ng/mL. Test rather than guess. Fat soluble and genuinely possible to overdo.

A probiotic, for quality of life rather than for antibodies

Based on the study that opens this guide.¹ *Lactiplantibacillus plantarum* 299v, added to nutritional change, improved quality of life substantially more than nutritional change alone, across 12 of 14 domains.

Be precise about the claim. Neither anti-TPO nor BMI changed. What changed was how people reported feeling, in a double-blind randomised design. That is a legitimate and under-valued outcome, and it is the specific thing most Hashimoto's supplement protocols never measure.

Caution: this is one strain in one study. Do not generalize it to probiotics as a category.

Zinc

Included because deficiency is common and produces overlapping symptoms, and because it is on the functional panel. Functional target 70 to 110 ug/dL. **Separate from levothyroxine by four hours.** Zinc taken long term can drive copper deficiency, so this is not one to take indefinitely without a reason.

Tier 3: Ask your practitioner first

I have left products and purchase links out of this tier on purpose. These are things worth raising with a clinician, and selling them from a page to someone I have never assessed would contradict the rest of this guide.

Myo-inositol combined with selenium. A 2026 systematic review with trial sequential analysis found myo-inositol plus selenium significantly reduced TSH compared with selenium alone in Hashimoto's with subclinical hypothyroidism, but found **no significant difference in T4 or in TPO antibodies.**⁶ The authors urge caution given the limited number of studies. A TSH-lowering supplement is a real thing to discuss with whoever manages your dose, not something to add quietly. *Ask:* "This lowered TSH in trials. Given I am on levothyroxine, does adding it risk over-replacement?"

T3 or desiccated thyroid, if you feel unwell on levothyroxine alone. This comes up constantly and deserves a straight answer. A meta-analysis of blinded randomized trials found that **approximately half of participants preferred combination T3 and T4 therapy over T4 alone, which was not distinguishable from chance.**⁷ A separate meta-analysis of 16 studies examined combination therapy and desiccated thyroid directly.⁸ In short, some people genuinely do better, the trials cannot yet tell us who, and it is a legitimate conversation rather than either a miracle or a fringe request. *Ask:* "I am biochemically controlled and still symptomatic. What is your view on a trial of combination therapy, and what would we measure?"

Anything at all, if you are pregnant or planning pregnancy. Requirements change substantially, iodine becomes a live question, and thyroid management in pregnancy is one of the few areas where treating subclinical hypothyroidism has stronger support.

Iodine or kelp. This is the most contested supplement in Hashimoto's and the diet guide in this kit works through it properly. The short version: excess iodine is implicated as a risk factor in this condition, and the antibody rise it can produce appears from population data to be a phase rather than a permanent state, with thyroid function minimally disrupted through it.^{9,10} I do use iodine clinically here, including kelp, which puts me at odds with most thyroid material and is a position I hold on clinical observation rather than trial evidence. If you use it, selenium goes with it, the dose needs to be known rather than variable, and your thyroid function gets rechecked. That is why it sits in this tier rather than lower down. *Ask:* "If I add iodine, what would you want to monitor, and how would my levothyroxine dose be affected?"

What I deliberately left out, and why

Anything sold as reversing Hashimoto's. Nothing has demonstrated that.

Antibody titers as a success metric. The probiotic study improved quality of life across 12 of 14 domains without moving anti-TPO.¹ The selenium trials moved anti-TPO without ever measuring quality of life.⁴ A protocol that judges itself on your antibody number has chosen the outcome that is easier to move, not the one you came for.

Proprietary thyroid support blends. They routinely contain iodine, often a lot, sometimes undisclosed. My objection is to not knowing the dose, since iodine itself is something I do use in this condition. Iodine in Hashimoto's is a decision worth making deliberately, and an undisclosed amount inside a blend takes that decision away from you.

Things I use in practice that are not in here. A guide sold to someone I have never examined comes without a history or a medication list, and it cannot change course next week. A document has to clear a higher bar than a conversation, and where the evidence does not support recommending something at a distance, it stays out. That is the reason you can trust what did make it in.

Running a fair trial

Timing first. Six weeks of consistent levothyroxine administration before you judge anything else.

One at a time. Selenium is the reasonable first, if you are going to trial anything.

Twelve weeks. The antibody studies ran three months; the probiotic study ran a comparable course.

Decide in advance what success means, and do not make it the antibody number. Energy, cognition, sleep, hair, cold tolerance, and how you feel by mid-afternoon are the outcomes worth tracking.

Track it: rootcauseunlocked.com/tracker

Where to find quality versions of these

Everything above was written without reference to any product line.

Third-party testing first. NSF Certified for Sport and USP Verified mean an independent lab confirmed contents. **For this condition specifically, also read the label for iodine**, since thyroid-support blends frequently include it and it is the one ingredient whose exact amount you most want to know.

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of sales through that link. Nothing here was included, excluded, ranked, or dosed based on what it earns. The single most-sold category in this condition, proprietary thyroid support blends, is in the do-not-buy section above.

When to seek clinical review

Urgently: if you have chest pain, a resting heart rate persistently over 120, severe confusion or drowsiness, or difficulty swallowing or breathing.

Promptly: if palpitations, tremor, heat intolerance, or unintended weight loss appear, since they can indicate over-replacement.

Before starting anything here: talk to your clinician first if you are pregnant, planning pregnancy, or taking multiple medications where timing becomes complicated.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

Iron, which is easy to miss and common in this population

Thyroid peroxidase is an iron-dependent enzyme, so iron status is not a side issue in thyroid function.

A systematic review and meta-analysis of iron deficiency and thyroid function found thyroid hormone levels lower in people with iron deficiency, particularly in pregnant women, and reported a possible relationship between iron deficiency, thyroid function, and autoimmunity in some patient groups (PMID 38004184). A separate review of the same kind, in reproductive-age and pregnant women, found significantly increased odds of overt and subclinical hypothyroidism in pregnant women with iron deficiency (PMID 33716980).

What to do. Test before supplementing. Ferritin with serum iron, TIBC, transferrin saturation, and a CBC. Iron is not a supplement to take speculatively: unnecessary iron causes constipation and nausea, overload is a genuine risk, and iron is a leading cause of poisoning death in young children, so keep it locked away.

Timing matters if you take levothyroxine. Iron interferes with its absorption. Separate them by at least four hours and tell your prescriber you are taking it.

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Functional target ranges transcribed from the Root Health blood chemistry range. Levothyroxine timing guidance reflects the product prescribing information.

For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

